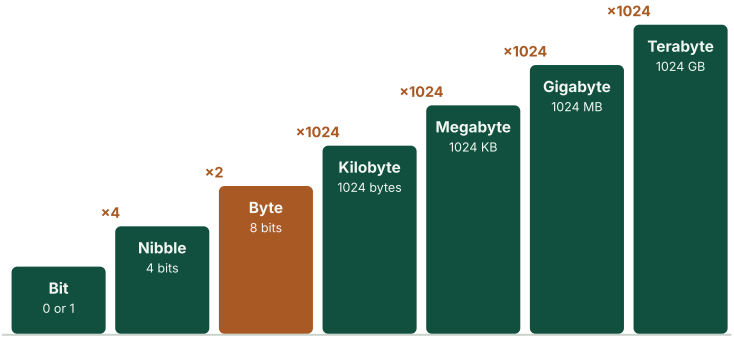
UNIT 1: DIGITAL TECHNOLOGY · DIGITAL DATA

Data, information and units of storage

L01 WORKSHEET

Name	Class	Date	Score

I can: Define data and information and explain how processing turns one into the other · Explain why computers store everything in binary and how ASCII represents characters · Order the units of storage from bit to terabyte · Convert between units of storage and work out how many files will fit on a device



Moving to a bigger unit: divide by 1024. Moving to a smaller unit: multiply by 1024.

Use this diagram to help you: Units of storage as a staircase from bit to terabyte, with the multiplier between each step

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Data**
- ☐ **Information**
- ☐ **Bit**
- ☐ **Nibble**
- ☐ **Byte**
- ☐ **Kilobyte (KB)**

A Data that has been processed and given context so that it has meaning.

B A group of 4 bits (half a byte). One hexadecimal digit can be stored in a nibble.

C 1024 bytes – enough for a short plain-text file.

D A binary digit: the smallest unit of digital data, either a 0 or a 1 (off or on).

E Raw facts and figures that have no meaning or context on their own.

F A group of 8 bits. One keyboard character (for example a letter, a digit or a punctuation mark) takes up 1 byte.

B Fill the gaps

Use the word bank · 8 marks

byte | information | nibble | data | ASCII | bit | context | 1024

Raw facts and figures with no meaning are called _____. When data is processed and given _____ it becomes _____. The smallest unit of storage is a _____, which is a single 0 or 1. A _____ is 4 bits and a _____ is 8 bits, enough to store one character using a character set such as _____. One megabyte is _____ kilobytes.

C True or false?

Tick one box · 9 marks

	True	False

A byte is 8 bits.	☐	☐
1 GB = 1024 KB.	☐	☐
'15, 22, 9' on its own is information.	☐	☐
A terabyte is larger than a gigabyte.	☐	☐
A nibble can store one hexadecimal digit.	☐	☐
To change bits into bytes you divide by 1024.	☐	☐
In ASCII, 'A' and 'a' have the same binary code.	☐	☐
One keyboard character is usually stored in 1 byte.	☐	☐
An information system takes data as input and produces information as output.	☐	☐

D Exam-style questions

27 marks

1 Using an example of each, describe the difference between data and information. [4]

2 Put these units in order from smallest to largest: megabyte, bit, terabyte, byte, nibble, kilobyte, gigabyte. [2]

3 (a) How many bits are in a nibble? (b) How many kilobytes are in a megabyte? (c) How many bytes are needed to store the word 'Fermanagh'? [3]

4 Place one of the following terms beside the correct definition. (Not all terms will be used.) BYTE, BIT, KILOBYTE. (a) The smallest unit of digital data, represented by either 0 or 1. (b) Describes eight binary digits. [2]

5 (a) A file is 6400 bits in size. How many bytes are required to store the file? (b) A school report is 6144 KB. What is its size in MB? Show your working. [3]

6 Convert 5 GB into MB. Show your working. [2]

7 Niamh saves a short poem as a plain-text file. The file is 5 KB. Calculate how many bits are needed to store the file. Show your working. [2]

8 A youth club in Omagh buys a 32 GB USB flash drive to store video clips. Each clip is 512 MB. Calculate how many clips the drive can hold. Show your working. [3]

9 Cian types a message on his laptop in Coleraine and emails it to a friend, whose tablet shows exactly the same text. Explain how a character set such as ASCII makes this possible. [2]

10 A GP surgery in Newry uses an information system to manage appointments. Describe how the system turns data into information, using examples from the surgery. [4]

E Quick quiz

Circle one answer · 9 marks

1 Which of these is information rather than data? [1]

A 14, 17, 21
B Q7X
C Sam scored 17 out of 20 in the maths test
D 01101

2 A phone has 64 GB of free space. Each photo is 4 MB. About how many photos will fit? [1]

A 16
B 256
C 16 384
D 65 536

3 How many bits make one byte? [1]

A 4
B 8
C 16
D 1024

4 Which unit is the largest? [1]

A Gigabyte
B Megabyte
C Kilobyte
D Terabyte

5 What is 2 MB in KB? [1]

A 2048 KB
B 1024 KB
C 16 KB
D 2000 KB

6 A file is 6400 bits. How many bytes is that? [1]

A 51 200 B 6.25

C 800 D 1600

7 Why does a computer use a character set such as ASCII? [1]

A To compress text files B To give each character a unique binary code

C To check spelling D To translate between languages

8 Roughly how much video would 1 GB hold? [1]

A About 20 minutes of HD video B About 1 second of HD video

C About 20 hours of HD video D About 1 minute of audio

9 How many bytes are needed to store the word 'Strabane'? [1]

A 8 B 7

C 64 D 16

UNIT 1: DIGITAL TECHNOLOGY · DIGITAL DATA

Data types

L02 WORKSHEET

Name	Class	Date	Score

I can: Name and describe the five data types in the specification and give an example of each · Explain why choosing the correct data type matters for storage, processing and validation · Choose the most suitable data type for an item of data and justify the choice · Recognise common date and time formats

Field	Example value	Data type
Surname	Maguire	String
Age	15	Integer
Height (m)	1.62	Real
Form initial	B	Character
Date of birth	12/09/2010	Date/time
Phone	07700 900123	String

Phone numbers are strings: the leading 0 must be kept and no maths is done on them.

Use this diagram to help you: A pupil record with each field matched to its data type

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Data type**
- ☐ **Integer**
- ☐ **Real**
- ☐ **Date/time**
- ☐ **Character**
- ☐ **String / text**

A Data type that stores dates and/or times so that they can be sorted, calculated with and shown in different formats.

B Data type for numbers that can have a decimal or fractional part, positive or negative. Real numbers also include whole numbers.

C The kind of data an item is (such as integer, real, date/time, character or string), which decides how it is stored and processed.

D Data type for a sequence of characters (letters, digits and symbols) of variable length.

E Data type for whole numbers, positive or negative, with no decimal or fractional part.

F Data type for a single letter, digit or symbol, stored using its code from the character set.

B Fill the gaps

Use the word bank · 7 marks

string | real | data type | character | integer | date/time | type check

Before data is processed it must be given a _____. A whole number such as 28 is stored as an _____, while a price such as 3.99 is stored as a _____ number. A single letter such as 'Y' is a _____, and a name or postcode is stored as a _____. A birthday is stored as

_____ so that it can be sorted and displayed in different formats. A _____ rejects data of the wrong kind.

C True or false?

Tick one box · 9 marks

	True	False
A postcode should be stored as a string.	☐	☐
Integers can store numbers with a decimal part.	☐	☐
Choosing the correct data type allows a type check to be used for validation.	☐	☐
A telephone number should be stored as an integer.	☐	☐
'double' is a numeric data type that can store a decimal number.	☐	☐
A score of 51 out of 100 is best stored as a string.	☐	☐
-61 is an integer.	☐	☐
A string must always contain the same number of characters.	☐	☐
12-Sep-2026 is an example of a medium date format.	☐	☐

D Exam-style questions

34 marks

1 State the most suitable data type for: (a) the number of pupils in a class (b) the price of a school lunch (c) a pupil's date of birth (d) a pupil's surname (e) a grade, e.g. A. [5]

2 Explain why a mobile phone number should be stored as a string and not an integer. [2]

3 Describe the difference between the integer and real data types. [2]

4 Give two reasons why data must be defined as a particular data type before it is processed. [2]

5 A student scores 51 in a Computing module. State the most appropriate data type for storing the student's score. [1]

6 Describe each of the following data types and give an example of data that would be stored using it: (a) numeric (integer or real/double) (b) date/time (c) text (character or string). [6]

7 From the list 4, 6.5, -3.2, 9, 15, 42.7: (a) identify all the integers (b) explain why all of the values could be stored as real numbers. [3]

8 State three different ways a date could be formatted when it is displayed by a computer, giving an example of each. [3]

9 A sports club in Ballymena stores these details for each member: membership number (e.g. 00417), first name, date joined, annual fee (e.g. £85.50), number of matches played and whether they have paid (Y or N). State the most suitable data type for each item and justify your choice for membership number and annual fee. [8]

10 A Year 9 pupil says 'The postcode BT74 6AA should be stored as an integer because it contains numbers.' Explain why the pupil is wrong. [2]

E Quick quiz

Circle one answer · 8 marks

1 Which data type is best for a postcode such as BT74 6AA? [1]

A Integer
B Real
C String
D Character

2 Why is a mobile phone number not stored as an integer? [1]

A It is too long to store
B Integers cannot be sorted
C It uses less storage
D The leading zero would be lost

3 Which data type should store a height of 1.75 m? [1]

A Real
B Integer
C Character
D Date/time

4 A field stores a single letter for shirt size (S, M or L). Which data type? [1]

A String
B Character
C Integer
D Real

5 Which item should be stored as an integer? [1]

A Price of a ticket
B Number of goals scored
C Phone number
D Team name

6 'int' and 'double' are both examples of which kind of data type? [1]

A Text
B Date/time
C Numeric
D Boolean

7 Which of these is a short date format? [1]

A Saturday, 12 September 2026
B 12-Sep-2026
C 15:45:30
D 12/09/2026

8 Which is NOT a reason for defining a data type before processing? [1]

A It makes the data more interesting to read
B It decides what processing can be done
C It allows a type check
D It helps the data be stored efficiently

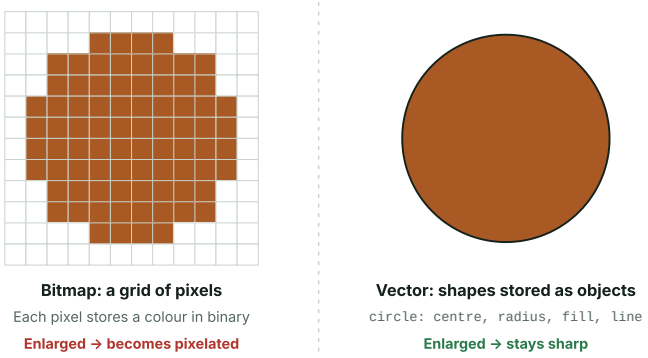
UNIT 1: DIGITAL TECHNOLOGY · DIGITAL DATA

Representing images: pixels, resolution, bitmap and vector

L03 WORKSHEET

Name	Class	Date	Score

I can: Explain how a bitmap image is made of pixels, each stored as a binary colour code · Explain how resolution and colour depth affect image quality and file size · Calculate the file size of a simple bitmap image · Compare bitmap and vector graphics, giving a suitable use for each



Use this diagram to help you: Bitmap image as a grid of pixels compared with a vector circle stored as an object

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Pixel**
- ☐ **Bitmap graphic**
- ☐ **Vector graphic**
- ☐ **Resolution**
- ☐ **Colour depth**
- ☐ **Pixelation**

A The blocky appearance of a bitmap image when it is enlarged too much, because the individual pixels become visible.

B The smallest element of a digital picture that a screen can show or an editor can change. Every pixel holds one colour, recorded as a binary code.

C An image stored as a list of objects (lines, curves and shapes) with properties such as position, size, line colour and fill colour. Can be enlarged without loss of quality. Example file type: SVG.

D The number of pixels in an image (width × height) or per inch. More pixels give a sharper image but a bigger file size.

E An image made of a grid of pixels, each of which can be a different colour. The colour code of every pixel is stored, plus metadata such as the dimensions. Pixelation occurs when it is enlarged. Example file type: JPEG.

F The number of bits used to store the colour of each pixel. n bits give 2^n colours.

B Fill the gaps

Use the word bank · 7 marks

colour depth | file size | pixels | vector | metadata | resolution | pixelated

A bitmap image is made of a grid of _____, and the colour of each one is stored as a binary code. The number of bits used per pixel is the _____. Increasing the _____ gives a sharper image but increases the _____. Data about the image, such as its width and height, is called _____. A _____ graphic is stored as objects and their properties, so it can be enlarged without loss of quality. Bitmap images become _____ when enlarged too much.

C True or false?

Tick one box · 9 marks

	True	False
Vector graphics lose quality when enlarged.	☐	☐
Photographs are best stored as bitmaps.	☐	☐
Higher resolution means a smaller file.	☐	☐
Vector graphics store shapes with properties such as fill colour.	☐	☐
A 24-bit colour image uses 24 bits to store each pixel.	☐	☐
A 2-bit image can use 4 different colours.	☐	☐
The colour codes alone are enough to rebuild a bitmap image.	☐	☐
An image with 5 colours needs only 2 bits per pixel.	☐	☐
SVG is an example of a vector file type.	☐	☐

D Exam-style questions

32 marks

1 Define the following terms: (a) pixel (b) image resolution. [2]

2 Explain how increasing the resolution of an image affects its quality and its file size. [3]

3 Describe how a vector graphic is stored. [2]

4 Describe two differences between bitmap and vector-based graphics. [4]

5 Ruairí takes a small photo of his team from the club website and stretches it to fill an A3 poster. The printed photo looks blocky. Explain why this happens. [2]

6 Describe what needs to be stored so that a bitmap image can be saved on a storage device and displayed again. [2]

7 A pupil designs a 10 × 10 pixel image using 4 colours. Calculate the minimum file size of the image data in bytes. Show your working. [3]

8 A photo for a school newsletter is 800 × 600 pixels and uses 24-bit colour. Calculate the size of the image data in kilobytes. Show your working. [3]

9 An image uses 16 different colours. State how many bits are needed to store each pixel and explain why. [2]

10 A designer is creating a company logo for a Derry bakery that will be printed on paper bags and on a large shop sign. Recommend a graphic type and justify your choice. [3]

11 A Belfast estate agent wants to put house photographs and a map of each area on its website. Discuss the use of bitmap and vector graphics for the website. In your answer, refer to how each is stored, quality and file size. [6]

E Quick quiz

Circle one answer · 9 marks

1 What happens when a bitmap image is enlarged a lot? [1]

A It stays sharp
B It becomes a vector
C It becomes pixelated
D The file size halves

2 Which image is best stored as a vector graphic? [1]

A A school logo
B A holiday photograph
C A scanned X-ray
D A webcam image

3 What does a bitmap image store for each pixel? [1]

A A shape
B A colour value in binary
C A line thickness
D A list of coordinates for curves

4 Increasing the resolution of an image will... [1]

A reduce the file size
B not change the file size
C turn it into a vector
D increase the file size

5 How many colours can a 2-bit colour image use? [1]

A 2
B 4
C 8
D 16

6 A 10 × 10 image uses 4 colours. What is the size of the image data? [1]

A 25 bytes
B 100 bytes
C 200 bytes
D 40 bytes

7 Which of these is metadata stored with a bitmap image? [1]

A The colour of pixel 1
B The image's width and height
C The fill colour of a circle
D The printer used

8 Which is a property stored in a vector graphic? [1]

A Fill colour of a shape
B Pixel colour depth
C Number of pixels per inch
D Compression ratio

9 Which file type is a vector graphic? [1]

A JPEG
B PNG
C GIF
D SVG

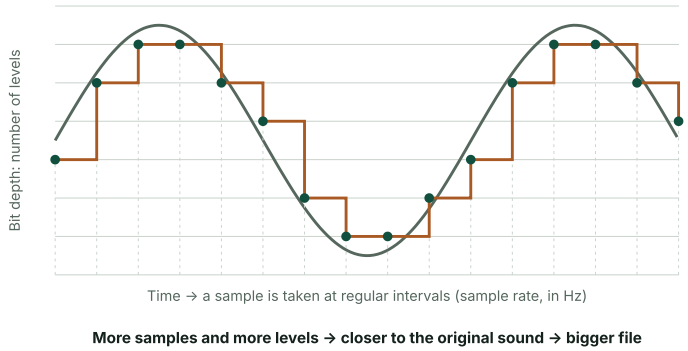
UNIT 1: DIGITAL TECHNOLOGY · DIGITAL DATA

Representing sound; buffering and streaming

L04 WORKSHEET

Name	Class	Date	Score

I can: Explain why sound must be converted from analogue to digital and describe what an ADC does · Describe how sample rate, bit depth and bit rate affect quality and file size · Calculate the bit rate and the file size of a simple sound recording · Explain how streaming and buffering let a video play while it downloads



Use this diagram to help you: A sound wave sampled at regular intervals, with sample rate on the time axis and bit depth as the number of levels

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Analogue signal**
- ☐ **Analogue-to-digital converter (ADC)**
- ☐ **Sampling**
- ☐ **Sample rate**
- ☐ **Bit depth**
- ☐ **Bit rate**

A The number of samples of a sound taken each second, measured in hertz (Hz). CD audio uses 44 100 Hz (44.1 kHz).

B A signal that varies smoothly and continuously, representing a physical quantity such as a sound wave.

C Measuring the level (amplitude) of an analogue signal at regular time intervals; each reading is rounded to a whole-number level and stored in binary.

D The number of bits of data processed for each second of sound, usually in kilobits per second; for one channel it equals sample rate × bit depth.

E A device that converts an analogue signal, such as sound from a microphone, into digital (binary) data by sampling it at regular intervals.

F How many bits are allocated to every sound sample. More bits give more possible levels, so each sample is recorded more accurately.

B Fill the gaps

Use the word bank · 8 marks

hertz · sample · bit depth · bit rate · ADC · sample rate · analogue · buffer

Sound is an _____ signal, so it must be converted to binary by an _____ before a computer can store it. Each measurement of the wave is called a _____. The number of measurements taken per second is the _____, measured in _____. The number of bits

allocated to every single measurement is the _____, and the number of bits processed per second is the _____. When streaming a video, the next part is stored temporarily in a _____ so playback carries on smoothly.

C True or false?

Tick one box · 9 marks

	True	False
A higher sample rate gives a smaller file.	☐	☐
Streaming lets media play before the whole file arrives.	☐	☐
Bit rate is measured in bits per second.	☐	☐
A buffer is a permanent storage device.	☐	☐
Standard CD audio takes 44 100 samples per second.	☐	☐
Each sample is rounded to the nearest whole-number level before it is stored as binary.	☐	☐
For one channel, bit rate = sample rate + bit depth.	☐	☐
A bit depth of 16 bits gives 65 536 possible values for each sample.	☐	☐
Doubling the length of an uncompressed recording halves its file size.	☐	☐

D Exam-style questions

32 marks

1 Explain why analogue-to-digital conversion is needed when recording sound on a computer. [2]

2 State the purpose of an analogue-to-digital converter (ADC). [1]

3 What is meant by sample rate? State the unit it is measured in. [2]

4 Describe how increasing the bit depth affects a sound recording. [2]

5 Distinguish between bit depth and bit rate. [2]

6 Standard CD audio is sampled at 44.1 kHz with a bit depth of 16 bits. Calculate the bit rate for one channel. Show your working. [2]

7 Orla records a 2-minute mono podcast intro for her school radio club at a sample rate of 22 050 Hz and a bit depth of 16 bits. Calculate the file size in bytes, and then in megabytes to two decimal places. Show your working. [4]

8 Explain how buffering allows a film to be streamed smoothly on a smart TV. [3]

9 Give one advantage and one disadvantage of streaming a film compared with downloading it. [2]

10 Explain how sound is recorded and stored on a computer. [6]

11 A music teacher in Omagh wants to record pupils' performances for their GCSE Music coursework. Discuss the choice of sample rate and bit depth for these recordings, referring to sound quality, file size and how the files will be stored and shared. [6]

E Quick quiz

Circle one answer · 8 marks

1 What does an analogue-to-digital converter (ADC) do when sound is recorded? [1]

A Converts digital data into sound for the speakers
B Compresses the sound file to make it smaller
C Samples the analogue signal and turns each sample into a binary number
D Stores the next part of a song in memory

2 What is the bit rate of a mono sound sampled at 8 000 Hz with an 8-bit depth? [1]

A 16 000 bps
B 64 000 bps
C 8 008 bps
D 1 000 bps

3 What is the unit of sample rate? [1]

A Hertz (Hz)
B Bits
C Bytes
D Pixels

4 Which change makes each sample more accurate? [1]

A Lower sample rate
B Fewer channels
C Lower bit rate
D Higher bit depth

5 Why is some detail lost when sound is sampled? [1]

A The microphone is digital
B What the wave does between samples is not recorded
C Binary cannot store numbers
D The buffer is too small

6 A 10-second stereo clip is recorded at 44 100 Hz with 16-bit samples. How many bits is it? [1]

A 705 600
B 1 411 200
C 7 056 000
D 14 112 000

7 What is a buffer? [1]

A A cable that speeds up downloads
B A compressed video format
C Memory that temporarily holds data before it is used
D A type of microphone

8 What happens when a streamed video stops and shows a loading icon? [1]

A The buffer has run out of data
B The file has finished downloading
C The bit depth is too high
D The ADC has stopped working

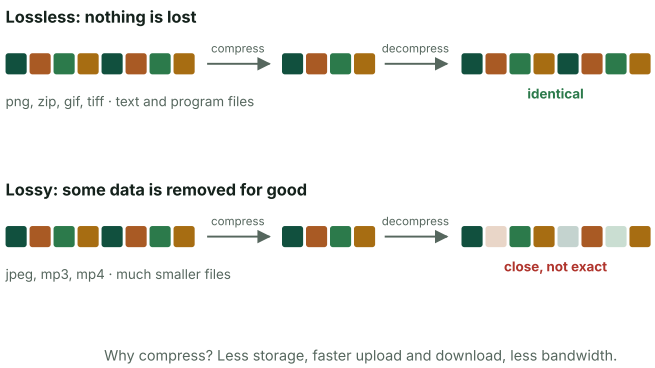
UNIT 1: DIGITAL TECHNOLOGY · DIGITAL DATA

Data portability, file formats and compression

L05 WORKSHEET

Name	Class	Date	Score

I can: Define data portability and explain why standard file formats support it · Describe the purpose and typical use of each of the 16 file formats in the specification · Choose a suitable file format for a task and justify the choice · Explain why data is compressed and compare lossy and lossless compression



Use this diagram to help you: Lossless compression restores every block of data; lossy compression permanently removes some

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Data portability**
- ☐ **File format**
- ☐ **Compression**
- ☐ **Lossy compression**
- ☐ **Lossless compression**
- ☐ **CSV**

A The standard way data is organised inside a file, shown by its file extension (e.g. .jpg, .csv), which determines which programs can open it.

B Compression that reduces file size by permanently removing some data, so some quality is lost and the original cannot be fully restored, e.g. jpeg, mp3.

C Comma-separated values: a text format in which each line is a record and the fields are separated by commas; used to move table data between spreadsheets and databases.

D Being able to move data between computers, devices or software applications without typing it in again, usually by saving it in a standard file format.

E Reducing the size of a file so it needs less storage and can be uploaded or downloaded more quickly.

F Compression that reduces file size without losing any data, so the file can be restored exactly as it was, e.g. zip, png.

B Fill the gaps

Use the word bank · 7 marks

pdf · lossless · portability · compression · csv · Lossy · midi

Moving data between applications without retyping it is called data _____. A _____ file stores table data with commas between the fields, while a _____ keeps its layout the same on any device. Making a file smaller is called _____. _____ compression permanently removes

some data, as in a jpeg photo, but _____ compression, such as zipping a file, lets the original be rebuilt exactly. The _____ format stores musical instructions rather than recorded sound.

C True or false?

Tick one box · 9 marks

	True	False
midi stores recorded sound.	☐	☐
jpeg uses lossy compression.	☐	☐
tiff is often used for high-quality printing.	☐	☐
Compressing a file increases the time it takes to upload.	☐	☐
Zipping a file with WinZip is an example of lossless compression.	☐	☐
wav is an uncompressed audio format.	☐	☐
wma was developed by Apple for iTunes.	☐	☐
Lossy compression is suitable for a spreadsheet of pupils' marks.	☐	☐
A pdf keeps the fonts and layout of a document the same on any device.	☐	☐

D Exam-style questions

38 marks

1 What is meant by data portability? [2]

2 Name a suitable file format for each: (a) a photo for a website (b) a document that must look the same on every device (c) data exported from a database to a spreadsheet (d) an animated image with few colours. [4]

3 Describe the difference between a midi file and an mp3 file. [2]

4 Give two reasons why data is compressed. [2]

5 Explain the difference between lossy and lossless compression. [2]

6 Explain why rtf may be chosen instead of a word processor's own file format. [2]

7 Explain why a user would compress a file before emailing it, and state which type of compression should be used for a spreadsheet of exam results. [3]

8 For each of the following file formats, state whether it is normally compressed or uncompressed: jpeg, wav, mp3, txt. [4]

9 A camera shop in Enniskillen takes photos of its stock. Compare jpeg and tiff as formats for (a) photos on the shop's website and (b) photos printed in a large poster for the window. [4]

10 Daniel needs to send his CV to several employers, who use different word processors and computers. Name two text or document file formats he could use and describe why each would suit him. [4]

11 A 3-minute song is stored as an uncompressed wav file of about 30 MB and as an mp3 of about 2.75 MB. Explain why the mp3 is smaller and describe one drawback of using the mp3. [3]

12 A local council in Northern Ireland wants to put a planning document, photographs of the sites and a table of planning decisions on its website for the public. Discuss suitable file formats for each type of data, referring to portability and compression. [6]

E Quick quiz

Circle one answer · 9 marks

1 What is meant by data portability? [1]

A Carrying a laptop between rooms

B Moving data between computers or applications without re-entering it

C Making a file as small as possible

D Storing data in the cloud

2 Which audio format stores musical instructions rather than recorded sound? [1]

A wav

B mp3

C wma

D midi

3 Which format stores data with each field separated by a comma? [1]

A rtf

B csv

C tiff

D wma

4 Which image format supports simple animation? [1]

A gif

B tiff

C jpeg

D pict

5 Lossy compression... [1]

A keeps every bit of the original

B always increases file size

C permanently removes some data

D is used for program files

6 Which format keeps a document's layout the same on any device? [1]

A txt

B csv

C midi

D pdf

7 Which format contains no formatting and can be opened by any word processor? [1]

A txt

B rtf

C pdf

D pict

8 Which format is best for a logo that needs a transparent background and no loss of quality? [1]

A jpeg

B mp4

C png

D csv

9 Which format was created by Microsoft to play audio and video in sync? [1]

A mp3

B avi

C midi

D wav

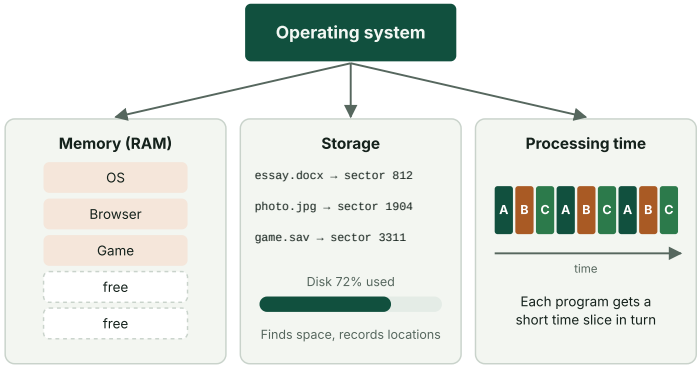
UNIT 1: DIGITAL TECHNOLOGY · SOFTWARE

System software and modes of processing

L06 WORKSHEET

Name	Class	Date	Score

I can: Distinguish between system software and application software, and between packaged and tailored software · Describe how the operating system allocates memory, storage and processing time · Describe real-time, batch and multi-user processing with examples · Describe how batch processing uses a master file and a transaction file to produce bills



Use this diagram to help you: The operating system allocating memory, storage and processing time

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **System software**
- ☐ **Application software**
- ☐ **Operating system**
- ☐ **Boot-up sequence**
- ☐ **Real-time processing**
- ☐ **Batch processing**

A The steps carried out when a computer is switched on: instructions in ROM check the hardware and load the operating system from storage into RAM.

B Software that runs and manages the computer hardware and acts as the interface between the hardware and application programs, e.g. the operating system, utilities and device drivers.

C Software designed to let the user carry out a specific task, such as word processing, spreadsheets or web browsing.

D Similar data is collected into batches over a period and processed together later, without human involvement, often overnight, e.g. bills and payroll.

E System software that manages the computer's resources – allocating memory, storage and processing time – and lets the hardware and application software work together.

F Each input is processed the moment it is entered, so the output is produced in time to affect the next input, e.g. ticket booking or air traffic control.

B Fill the gaps

Use the word bank · 8 marks

time slice · real-time · batch · operating system · master file · transaction file · ROM · multi-user

The _____ is system software that allocates memory, storage and processing time. It gives each task a _____ of the processor in turn. When a computer is switched on, instructions stored in _____ check the hardware and load the operating system. Ticket booking systems use _____ processing so that seats are updated immediately. Electricity bills are produced by _____ processing, which matches a temporary _____ against the permanent _____. Many users sharing one processor using the round robin method is a _____ system.

C True or false?

Tick one box · 9 marks

	True	False
Batch processing needs a person to check each record as it is processed.	☐	☐
Time slicing gives each program a share of the processor's time.	☐	☐
The boot-up instructions are stored in RAM.	☐	☐
Airline bookings use real-time processing.	☐	☐
A device driver is system software.	☐	☐
In batch processing the transaction file is kept permanently after processing.	☐	☐
In a multi-user system the OS uses a round robin approach to share the processor.	☐	☐
Tailored software is usually cheaper than packaged software.	☐	☐
Higher-priority tasks can be given more time slices.	☐	☐

D Exam-style questions

48 marks

1 Describe the difference between system software and application software. [2]

2 Classify each software example as either system software (S) or application software (A): Windows 11 operating system; Microsoft Excel spreadsheet; patient record database; disk defragmenter software; device driver; Google Chrome web browser. [6]

3 Describe two differences between packaged software and tailored software. [4]

4 Describe how system software manages memory. [2]

5 Describe how system software manages storage. [2]

6 Describe how system software allocates processing time. [2]

7 Describe what happens during the boot-up sequence. [3]

8 Omar runs a small business in Newry manufacturing parts for bikes. His staff have their own computer systems. (a) What do the letters CPU stand for in relation to a computer system? (b) List two resources managed by the operating system in a computer system. [3]

9 The Grand Opera House in Belfast sells theatre tickets online. Describe the main features of the real-time processing its booking system uses. [4]

10 A district council in Northern Ireland pays its staff once a month. Describe the main features of batch processing as used for the council's payroll. [4]

11 An electricity company uses batch processing to produce customer bills. Describe how the bills are produced, referring to the master file and the transaction file. [5]

12 Describe how a multi-user system allows many users to share one processor. [3]

13 Explain the difference between real-time processing and multi-user processing. [2]

14 A leisure centre in Derry~Londonderry wants a computer system for (i) online booking of badminton courts and (ii) paying its staff each month. Discuss which mode of processing should be used for each task, justifying your choices. [6]

E Quick quiz

Circle one answer · 9 marks

1 An airline booking system must mark a seat as sold the moment it is booked. Which mode of processing is this? [1]

A Batch processing
B Real-time processing
C Lossy processing
D Offline processing

2 Which task is best suited to batch processing? [1]

A Booking a concert seat
B Controlling traffic lights
C Producing monthly electricity bills
D Playing an online game

3 Which of these is system software? [1]

A A spreadsheet
B A web browser
C A photo editor
D An operating system

4 What does the OS do when allocating processing time? [1]

A Gives each task a time slice of the processor
B Defragments the disk

C Encrypts files
D Backs up data

5 Where are the first instructions run at boot-up stored? [1]

A RAM
B Cache
C The cloud
D ROM

6 In batch processing, which file is permanent or semi-permanent? [1]

A Master file
B Transaction file
C Buffer
D Temporary file

7 A school's pupil-records database written just for it is... [1]

A packaged software
B tailored software
C utility software
D freeware

8 In a multi-user system, how is the processor shared? [1]

A Users queue until the first user logs off
B Each user gets a time slice in turn (round robin)
C Each user gets their own processor
D Jobs are saved and run overnight

9 What does CPU stand for? [1]

A Computer Personal Unit
B Central Program Utility
C Central Processing Unit
D Control Processing Unit

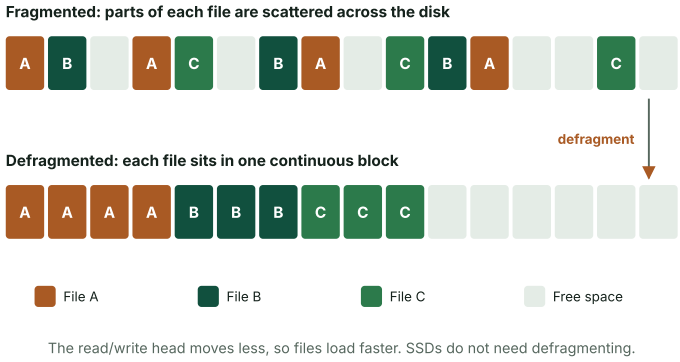
UNIT 1: DIGITAL TECHNOLOGY · SOFTWARE

Utility applications and antivirus software

L07 WORKSHEET

Name	Class	Date	Score

I can: Explain why files become fragmented and how a defragmenter speeds up file access · Describe what a task scheduler does, including round robin sharing of processor time · Describe full and incremental backups, how data is restored and why backups are needed · Explain how antivirus software works and why it must be updated regularly



The read/write head moves less, so files load faster. SSDs do not need defragmenting.

Use this diagram to help you: Disk blocks before and after defragmentation

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Utility software**
- ☐ **File fragmentation**
- ☐ **Disk defragmenter**
- ☐ **Task scheduler**
- ☐ **Round robin**
- ☐ **Backup**

A A utility that rearranges the fragments of files on a disk so each file is stored in continuous blocks and free space is grouped together, speeding up file access.

B A program that carries out one specific task to help the operating system maintain, protect or speed up the computer, e.g. defragmenter, task scheduler, backup, antivirus.

C A copy of data stored separately, preferably off-site, so it can be restored if the original is lost or damaged.

D When a file's data blocks are stored in separate, non-adjacent places on a disk because there is not enough continuous free space.

E A utility that shares processor time between tasks (e.g. round robin) and runs jobs such as backups, virus scans or updates automatically at set times.

F A scheduling method in which each task or user is given a time slice of the processor in turn, returning to the first after the last.

B Fill the gaps

Use the word bank · 7 marks

signatures · task scheduler · incremental backup · quarantine · backup · disk defragmenter · updated

A _____ moves the scattered blocks of each file together so it can be read faster. A _____ can run jobs such as backups automatically at set times. A copy of data kept separately in case the original is lost is a _____. An _____ only copies data that has changed since the last

backup. Antivirus software compares files with a database of virus _____ and moves infected files into _____. It must be _____ regularly so that new viruses can be detected.

C True or false?

Tick one box · 9 marks

	True	False
Defragmenting can improve the speed of a hard disk drive.	☐	☐
Antivirus software never needs updating once it is installed.	☐	☐
Restoring means copying data back from a backup.	☐	☐
A task scheduler only runs a job when the user clicks on it.	☐	☐
WinZip is an example of utility software.	☐	☐
Defragmentation puts the parts of each file back into continuous blocks.	☐	☐
A full backup only copies files that have changed.	☐	☐
Backups should be stored in the same room as the server for convenience.	☐	☐
Antivirus software can move an infected file into quarantine so it cannot run.	☐	☐

D Exam-style questions

38 marks

1 The operating system on John's computer includes a utility application. What is the function of the following two tasks carried out by utility software? (i) Disk defragmenting (ii) Task scheduling [4]

2 Explain why files become fragmented on a hard disk drive. [3]

3 Describe how disk defragmenting improves the performance of a computer. [3]

4 Explain why a solid-state drive should not be defragmented. [2]

5 Describe how the round robin method is used to share a processor between several users. [3]

6 Give two examples of tasks that could be scheduled to run automatically. [2]

7 Why do we need to back up our data? [3]

8 Describe the difference between a full and an incremental backup. [2]

9 State three places where data can be backed up to. [3]

10 Describe how antivirus software detects and deals with a virus. [3]

11 Explain why antivirus software must be updated regularly. [2]

12 Explain why it is important to keep software up to date. [2]

13 A GP surgery in Ballymena stores all its patient records on one server. Discuss a suitable backup and restore procedure for the surgery, referring to the type of backup, how often it should happen, where the backup should be stored and how utility software can help. [6]

E Quick quiz

Circle one answer · 8 marks

1 Why does a disk defragmenter make a hard disk drive faster? [1]

A It deletes unwanted files

B It stores each file's blocks together so the read/write head moves less

C It compresses every file

D It scans for viruses

2 Why must antivirus software be updated regularly? [1]

A To make the computer start faster

B To free up disk space

C So it can recognise the signatures of newly created viruses

D So it can defragment the disk

3 An incremental backup copies... [1]

A only files changed since the last backup

B everything on the drive

C only the operating system

D only photos

4 Which utility would run a backup automatically every night? [1]

A Defragmenter

B Firewall

C Compiler

D Task scheduler

5 After defragmenting, the blocks A1 B3 A2 A4 B2 A3 B1 B4 become... [1]

A A1 B1 A2 B2 A3 B3 A4 B4

B B1 B2 B3 B4 A1 A2 A3 A4

C A1 A2 A3 A4 B1 B2 B3 B4

D Unchanged

6 Which is NOT a reason to back up data? [1]

A The hard drive may fail

B A virus may destroy the data

C A fire may destroy the building

D It makes the processor faster

7 In round robin scheduling, what happens after the processor serves the last computer? [1]

A It switches off

B It starts again with the first computer

C It waits for the user to log off

D It serves the busiest computer only

8 What is a virus signature? [1]

A A pattern of code that identifies a particular virus

B The name of the person who wrote the virus

C A password to unlock infected files

D A digital certificate on a website

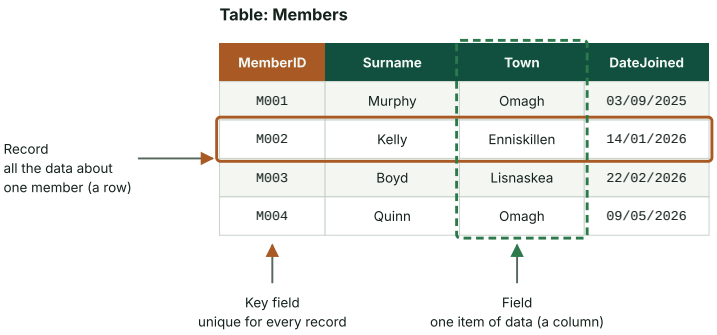
UNIT 1: DIGITAL TECHNOLOGY · DATABASE APPLICATIONS

Database concepts

L08 WORKSHEET

Name	Class	Date	Score

I can: Define table, record, field and key field and identify them in a given table · Explain why a key field must be unique and choose a suitable key field · Explain how tables are linked by a relationship using a primary key and a foreign key · Describe what queries, forms, reports and macros are used for · Explain what importing data means and why it is useful



Use this diagram to help you: A database table showing a record, a field and the key field

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Database**
- ☐ **Table**
- ☐ **Flat file**
- ☐ **Record**
- ☐ **Field**
- ☐ **Key field / primary key**

A A database that holds all its data in a single two-dimensional table, which often leads to repeated data.

B An organised, structured collection of related data that can be searched, sorted and presented, usually held in one or more tables.

C A database structure that stores data about one type of thing, organised in rows (records) and columns (fields).

D A field whose value is unique for every record, used to identify each record in a table.

E All of the related data about one object or person, forming a complete row in a table. A record is made up of fields.

F A single item of data stored about every object or person, forming a column in a table and identified by its field name.

B Fill the gaps

Use the word bank · 8 marks

field · key field · record · relationship · query · tables · foreign key · report

A database stores data in _____. Each row is a _____, which holds all the data about one item, and each column is a _____. The _____ uniquely identifies each record. When the primary key of one table is stored in another table it is called a _____, and this creates a

_____. A _____ searches for records that match criteria, and a _____ presents the results in a formatted layout.

C True or false?

Tick one box · 9 marks

	True	False
A field is a row in a table.	☐	☐
A key field must contain a unique value for every record.	☐	☐
Forms are used to enter, edit and view records.	☐	☐
A relational database stores all its data in one table.	☐	☐
A composite key is a primary key made up of two or more fields.	☐	☐
A foreign key must be unique in the table it appears in.	☐	☐
Enforcing referential integrity stops a record being linked to a parent record that does not exist.	☐	☐
A report is used to search the database for records that match criteria.	☐	☐
Importing data from a csv file saves time and avoids retyping errors.	☐	☐

D Exam-style questions

34 marks

1 Define the terms (a) record and (b) field. [2]

2 Look at the Erne Valley Swimming Club Members table on the slides. (a) State how many records the table contains. (b) Identify the key field and justify your choice. (c) Explain why Surname would not be a suitable key field. [4]

3 Describe the purpose of a form in a database, and give two ways a form can help to keep the data accurate. [3]

4 A leisure centre in Newry wants a printed list of all members whose membership ends this month, grouped by membership type. Explain how a query and a report would each be used to produce this. [4]

5 The receptionist at the leisure centre adds new members and prints a welcome letter many times each day. Explain one benefit of using a macro for this task. [2]

6 A swimming club in Lisburn has 200 new members' details in a spreadsheet from its online sign-up form. Explain what importing this data into the club database would involve, and give one benefit of importing rather than typing it in. [2]

7 Explain what is meant by a foreign key. Use a Doctor table and a Patient table as an example. [2]

8 One doctor has many patients, but each patient has only one doctor. State the type of relationship between the Doctor and Patient tables and explain one benefit of enforcing referential integrity on it. [3]

9 What is meant by a composite key? Give an example. [2]

10 A youth club in Strabane stores all its data in one flat file. Each time a member books an event, their name, address and phone number are typed again. Explain two advantages of using a relational database instead. [4]

11 A school library in Omagh is designing a database with three tables: Pupils, Books and Loans. Discuss how the database should be designed, referring to tables, key fields, relationships and the database tools the librarian would use. [6]

E Quick quiz

Circle one answer · 8 marks

1 In a database table, what is one row called? [1]

A A field
B A record
C A query
D A macro

2 A swimming club stores Surname, Squad, DateOfBirth and MemberID. Which is the best key field? [1]

A MemberID
B Surname
C Squad
D DateOfBirth

3 What is a query used for? [1]

A Entering data one record at a time
B Printing a formatted summary
C Searching for records that match criteria
D Recording keystrokes

4 A set of stored instructions that runs a database task automatically when a button is clicked is a... [1]

A form
B relationship
C field
D macro

5 When the primary key of one table appears as a field in a second table, it is called a... [1]

A composite key
B foreign key
C lookup
D macro

6 One doctor has many patients; each patient has one doctor. Where does the ∞ or crow's foot go? [1]

A At the Doctor end
B At both ends
C At the Patient end
D Nowhere – it is one-to-one

7 What is the main problem with storing everything in a flat file? [1]

A Data is repeated, wasting space and causing inconsistencies
B It cannot be printed
C It cannot be sorted
D It cannot have a key field

8 Which tool would present club members grouped by squad, ready to print? [1]

A Form
B Macro
C Relationship
D Report

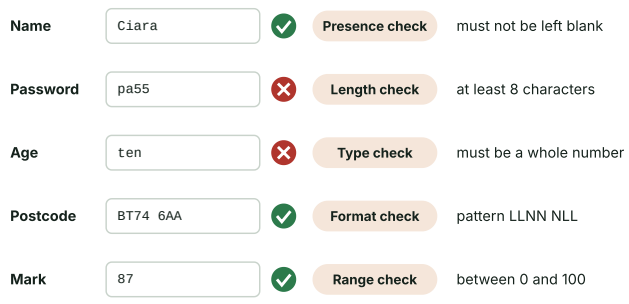
UNIT 1: DIGITAL TECHNOLOGY · DATABASE APPLICATIONS

Database data types and validation

L09 WORKSHEET

Name	Class	Date	Score

I can: Choose and justify a suitable data type for each field in a table · Explain why validation is needed and what it cannot do · Describe presence, length, type, format and range checks with examples · Write validation rules and input masks for database fields · Explain the difference between validation and verification



Use this diagram to help you: Five validation checks applied to form fields, with passes and failures

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Data type**
- ☐ **Validation**
- ☐ **Presence check**
- ☐ **Length check**
- ☐ **Type check**
- ☐ **Format check**

A A check that makes sure the data entered has an exact number of characters, or is no shorter or longer than set limits.

B A check that makes sure data has been entered in a required field, so the field cannot be left blank.

C The kind of data a field can hold, e.g. Short Text, Number, Currency or Date/Time. It controls what can be entered and what can be done with the data.

D An automatic check made by the computer as data is input, to make sure the data is sensible, reasonable and inside allowed limits. Data that breaks the rules is rejected; validation cannot confirm the data is correct.

E A check that makes sure the data entered matches a set pattern, e.g. a postcode or an email address containing @.

F A check that makes sure the data entered is of the correct data type, e.g. numeric, text or date.

B Fill the gaps

Use the word bank · 8 marks

format | presence | Date/Time | data type | range | length | Validation | correct

Each field in a table is given a _____, such as Currency for money or _____ for a booking date. _____ is an automatic check that data is sensible. A _____ check makes sure a field is not left blank. A _____ check counts the number of characters. A _____ check makes

sure data matches a pattern such as a postcode. A _____ check makes sure a value is between two limits. Validation cannot tell whether data is _____.

C True or false?

Tick one box · 9 marks

	True	False
Validation proves that data is accurate.	☐	☐
A range check suits a field that stores an exam mark from 0 to 100.	☐	☐
A format check suits a postcode.	☐	☐
A type check counts the number of characters in a field.	☐	☐
A telephone number should be stored with the Number data type.	☐	☐
The Currency data type stores money values with two decimal places.	☐	☐
The Lookup Wizard creates a list of values that can be picked from a drop-down list.	☐	☐
In Access, Len([Ability])>7 is an example of a range check.	☐	☐
Typing an email address twice so the computer can compare them is a form of verification.	☐	☐

D Exam-style questions

43 marks

1 Why is validation used when data is entered into a database? [2]

2 Explain why validation cannot guarantee that data is correct. Use an example. [2]

3 Describe a range check and give an example of a field where one would be used. [2]

4 Name and describe a suitable validation check for each field: (a) Postcode (b) Surname (c) Number of tickets (maximum 10). [6]

5 Describe the difference between a length check and a format check. [2]

6 Describe how a presence check can be set up for a field in a database table. [2]

7 A Film Characters table has a field called Ability which must be at least 8 characters long. Write a validation rule for this field and name the type of check. [2]

8 A group must have a minimum of 5 and a maximum of 20 members. Write a validation rule for the GroupSize field and name the type of check. [2]

9 State the most suitable data type for each field: (a) a URL for a character's fan page (b) a photo stored with each record (c) the price of a cinema ticket (d) a unique number generated automatically for each new record. [4]

10 A youth football club in Dungannon stores PlayerPhone, DateOfBirth, ShirtNumber and SubsPaid for each player. State the most suitable data type for each field and justify each choice. [8]

11 A school canteen in Coleraine records how much money each pupil tops up, from £1 to £50. Write a validation rule for the TopUp field and suitable validation text. [2]

12 Explain the difference between validation and verification. Describe one method of verification. [3]

13 A GP surgery in Ballymena is setting up a database to book patient appointments. Discuss how data types and validation checks can be used to make the appointment data as accurate as possible, and explain the limits of validation. [6]

E Quick quiz

Circle one answer · 8 marks

1 Which validation check makes sure a field is not left empty? [1]

A Range check
B Presence check
C Format check
D Type check

2 Which data type is best for a monthly club fee of £24.99? [1]

A Currency
B Number (integer)
C Short Text
D Date/Time

3 Checking that an email address contains an @ symbol is a... [1]

A type check
B range check
C format check
D presence check

4 Which statement about validation is true? [1]

A It guarantees data is correct
B It is carried out by a person reading the data
C It encrypts the data
D It cannot tell if data is correct, only if it is sensible

5 Which validation rule accepts a group of 5 to 20 people, including 5 and 20? [1]

A >5 And <20
B <5 Or >20
C >=5 And <=20
D Len([GroupSize])>4

6 Which data type stores a file made in other software, such as a spreadsheet, using object linking and embedding? [1]

A Hyperlink
B Calculated
C Lookup Wizard
D OLE Object

7 Which data type automatically creates a new sequential number for each record? [1]

A AutoNumber
B Number
C Yes/No
D Short Text

8 In an Access input mask, what does the character L mean? [1]

A A digit is required
B A letter is required
C A lower-case letter is optional
D Any character is allowed

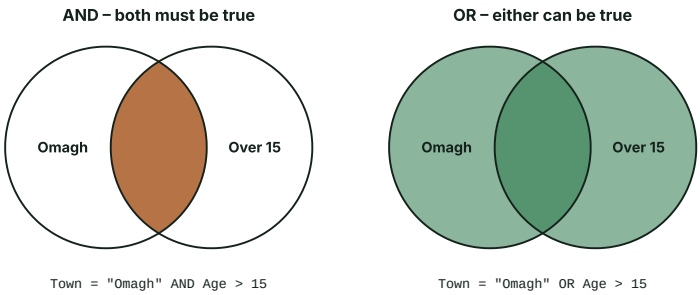
UNIT 1: DIGITAL TECHNOLOGY · DATABASE APPLICATIONS

Queries, logical operators and big data

L10 WORKSHEET

Name	Class	Date	Score

I can: Write query criteria using comparison operators and AND, OR and BETWEEN · Set up a query in a query design grid and predict which records it returns · Explain big data using volume, velocity and variety · Tell structured and unstructured data apart · Explain why data analytics is needed to make use of big data



`Town = "Omagh" AND Age > 15`
narrows the results

`Town = "Omagh" OR Age > 15`
widens the results

BETWEEN 10 AND 20 includes both 10 and 20.

Use this diagram to help you: Venn diagrams: AND returns records in both sets, OR returns records in either

A Key terms

`Write the letter of the matching definition · 6 marks`

- ☐ **Query**
- ☐ **Criteria**
- ☐ **Comparison operator**
- ☐ **Logical operator**
- ☐ **BETWEEN**
- ☐ **Group by**

A The conditions a record must meet to be included in the results of a query, e.g. Age >= 18.

B A way of searching a database to extract the records that meet set criteria. It checks each record and lists the data that satisfies the criteria.

C Organises query or report results by the values of a field, e.g. grouping members by squad.

D A symbol used to compare a value with other data, giving true or false: < less than, <= less than or equal to, > greater than, >= greater than or equal to, = equal to, <> not equal to.

E An operator such as AND, OR or BETWEEN that combines or tests conditions and gives a true or false (Boolean) result.

F A logical operator that selects values within a range, including both end values.

B Fill the gaps

`Use the word bank · 8 marks`

variety | OR | BETWEEN | query | AND | velocity | criteria | analytics

A _____ extracts records that meet set _____. The _____ operator needs both conditions to be true, while _____ needs at least one to be true. _____ finds values within a range, including both ends. Big data is described by volume, _____ and _____. Using software to find patterns and trends in big data is called data _____.

C True or false?

Tick one box · 9 marks

	True	False
OR returns records where at least one of the conditions is true.	☐	☐
Velocity refers to the amount of data stored.	☐	☐
BETWEEN 10 AND 20 includes the value 20.	☐	☐
Big data is easy to analyse by hand.	☐	☐
The comparison operator <> means 'not equal to'.	☐	☐
The criteria Goals > 8 returns a player who has scored exactly 8 goals.	☐	☐
In a query design grid, criteria typed on the same row are joined with AND.	☐	☐
Social media posts and essays are examples of structured data.	☐	☐
Data mining software is used to find hidden patterns in very large data sets.	☐	☐

D Exam-style questions

34 marks

1 Write the criteria to find all members who are aged 18 or over. [1]

2 Write the criteria to find all bookings made between 1 March 2026 and 31 March 2026. [2]

3 Explain the difference between the AND and OR operators in a query. [2]

4 Describe 'big data' by making reference to volume, velocity and variety. [6]

5 Explain why data analytics is needed to interpret big data. [2]

6 Describe the difference between structured and unstructured data. Give an example of each. [4]

7 Using the Fermanagh Junior Football League Players table on the slides, write the criteria to find every player who is not in the Enniskillen club, and state the PlayerIDs it returns. [2]

8 Using the Fermanagh Junior Football League Players table on the slides, state the PlayerIDs returned by each query: (a) Goals < 9 (b) Club = "Enniskillen" AND Age = 14 (c) Age = 13 OR Goals >= 12. [3]

9 A garden centre in Lisburn stores products in a table with the fields ProductName, Category, Price and InStock. Write criteria to find all products in the category "Plants" that cost less than £20. [2]

10 The garden centre manager wants a list of products costing from £10 to £25 inclusive, sorted by price, showing only ProductName and Price. Describe how the query should be set up in a query design grid. [4]

11 A supermarket chain in Northern Ireland collects data from loyalty cards, self-service tills, its app and social media. Discuss how this is an example of big data and how data analytics could help the business. [6]

E Quick quiz

Circle one answer · 8 marks

1 Which operator finds values from 10 to 20, including 10 and 20? [1]

A OR
B <
C BETWEEN
D <>

2 Which of the three Vs of big data describes the speed at which data is created and processed? [1]

A Velocity
B Volume
C Variety
D Validation

3 Town = "Omagh" AND Age > 15 returns records where... [1]

A either condition is true
B neither condition is true
C only Town is checked
D both conditions are true

4 Social media posts, videos and GPS data collected together are an example of... [1]

A volume
B variety
C velocity
D validation

5 Does Score >= 50 return a record with a score of exactly 50? [1]

A Yes
B No
C Only with OR
D Only with BETWEEN

6 Which criteria finds every player except those in the Senior squad? [1]

A Squad = "Senior"
B Squad > "Senior"
C Squad <> "Senior"
D Squad BETWEEN "Senior"

7 Which of these is unstructured data? [1]

A A table of dates of birth
B Customer addresses in a database
C A list of exam marks
D Video clips and social media posts

8 What is data analytics? [1]

A Typing data into a form
B Backing up a database
C Using software to find patterns, trends and relationships in data
D Checking data as it is entered

UNIT 1: DIGITAL TECHNOLOGY · SPREADSHEET APPLICATIONS

Spreadsheets 1: structure, formatting, formulae and cell referencing

L11 WORKSHEET

Name	Class	Date	Score

I can: Identify cells, rows, columns, ranges and worksheets, and give a cell reference correctly · Choose a suitable data format for a column and describe formatting that improves presentation · Write formulae and simple functions using the correct symbols and range syntax · Explain and use relative and absolute cell references when replicating formulae



Use this diagram to help you: Spreadsheet formulas copied down: relative references change, the absolute reference stays fixed

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Spreadsheet**
- ☐ **Cell range**
- ☐ **Label**
- ☐ **Value**
- ☐ **Formula**
- ☐ **Spreadsheet function**

A A built-in formula that does a specific job, e.g. SUM, AVERAGE, MAX, MIN, COUNT, IF and VLOOKUP.

B Application software that organises data in a grid of rows and columns so it can be calculated, analysed and presented. A spreadsheet file (workbook) contains one or more worksheets.

C Text used to identify the contents of a row or column, e.g. Name, Age.

D A block of adjacent cells referred to together, written first cell:last cell, e.g. B5:B12.

E A number entered into a spreadsheet so that calculations can be carried out, e.g. 5, 23.

F An instruction in a cell to carry out a calculation using operators and cell references, e.g. =A1+B1. It always starts with = and the cell shows the result, which updates when the referenced cells change.

B Fill the gaps

Use the word bank · 8 marks

cell · MIN · range · SUM · worksheets · formula · replication · absolute

A spreadsheet can contain several _____, each made of rows and columns. A _____ is identified by its column letter and row number, such as C4, and a block of cells such as B2:B10 is called a _____. Every _____ starts with an equals sign. The _____ function adds the values

in a range and _____ returns the smallest value. Copying a formula down a column is called _____. A reference written with dollar signs, such as $F$1, is an _____ reference because it does not change when copied.

C True or false?

Tick one box · 9 marks

	True	False
$A$1 is an absolute reference.	☐	☐
Formulae use x for multiply.	☐	☐
A spreadsheet can contain more than one worksheet.	☐	☐
A label is a number entered so that calculations can be carried out.	☐	☐
=SUM(A1:A10) adds A1 and A10 only.	☐	☐
When a formula is replicated, relative references change to match the row being used.	☐	☐
The fill handle is the small square at the bottom right of the selected cell.	☐	☐
COUNT counts every cell in a range that is not empty.	☐	☐
Merge and centre combines selected cells into one cell and centres the contents.	☐	☐

D Exam-style questions

48 marks

1 Describe the difference between a row and a column in a spreadsheet. [2]

2 A sports club records match fees in cells B2 to B12. Write a function to add the values in B2 to B12. [1]

3 Write a function to find the highest value in cells C2 to C20. [1]

4 Write a function to find the lowest value in cells C2 to C20. [1]

5 Write a function to calculate the arithmetic mean (average) of the values in A1 to A10. [1]

6 Write a formula for cell B7 that adds the cells B1 to B6 together. [1]

7 State the 'golden rule' for entering a formula and write a formula to divide the contents of A3 by the contents of A4. [2]

8 A youth club budget sheet holds the money available in I1 and the money spent so far in I2. Write a formula for I3 to show how much money is left. If I1 is £640.00 and I2 is £415.50, state the value shown in I3. [2]

9 A Bangor sailing club records course bookings in a sheet with the columns Booked By, Course, Places, Price Per Place, Member Discount and Amount Due. State the most suitable data format for (a) Booked By, (b) Price Per Place, (c) Member Discount. [3]

10 Cell E2 contains =C2*D2. State the formula in E3 after E2 is replicated into E3. [1]

11 Cell C1 contains =A1*B1. The formula is replicated down to C16 using the fill handle. (a) State the formula in C9. (b) Name the type of cell referencing used. [2]

12 Describe what is meant by replicating formulae and give one benefit. [2]

13 Explain why an absolute cell reference would be used for a VAT rate stored in cell H1. [2]

14 In a bakery's order sheet, cell E2 contains =D2*(1-$H$1) and is replicated down to E6. Cell H1 holds the discount rate. Explain why $H$1 has been used. [2]

15 Explain the difference between relative and absolute cell referencing. Use an example of each. [4]

16 A school tuck shop sheet has takings of £27.00, £24.00, £16.50, £39.60 and £13.20 in cells D2 to D6. State the result of (a) =SUM(D2:D6), (b) =MAX(D2:D6), (c) =MIN(D2:D6). [3]

17 A teacher has entered test scores in D2 to D30, but some pupils were absent and their cells are blank. Write a function to count how many pupils have a score. [1]

18 Column C of a camp sign-up sheet (C2 to C40) shows "Yes" if the medical form has been returned. Write a function to count how many forms have been returned. [2]

19 Explain why a formula such as =B2*C2 should be used in a cell instead of typing in the answer. [2]

20 Give one reason for selecting a print area. [1]

21 Describe what merge and centre does and give an example of when it would be used. [2]

22 Describe two features that could be used to improve the presentation of a spreadsheet that will be printed. [4]

23 A Fermanagh football club records its members' details, subscriptions and payments on paper. Discuss the benefits and drawbacks of moving these records to a spreadsheet. In your answer refer to formulae, formatting and printing. [6]

E Quick quiz

Circle one answer · 9 marks

1 Which cell reference identifies column D, row 7? [1]

A 7D B D7

C D:7 D 7D

2 In a tuck shop sheet, which function finds the smallest value in D2 to D6? [1]

A =SMALL(D2:D6) B =MIN(D2:D6)

C =LOW(D2:D6) D =MIN(D2,D6)

3 Every formula must start with... [1]

A + B #

C = D $

4 Which is an absolute reference? [1]

A B2 B B$

C #B2 D $B$2

5 =B4/C4 replicated to row 5 becomes... [1]

A =B5/C5 B =B4/C4

C =C4/D4 D =$B$5/$C$5

6 A cell formatted as Percentage holds 0.2. What does it show? [1]

A 0.2 B 0.2%

C £0.20 D 20%

7 =D1-C1 does what? [1]

A Subtracts D1 from C1 B Adds C1 and D1

C Subtracts C1 from D1 D Finds the difference between columns C and D

8 E2 contains =D2*(1-H1) and is filled down. What is in E3? [1]

A =D3*(1-$H$1) B =D2*(1-H2)

C =D3*(1-H1) D =D3*(1-H2)

9 Cells D2:D6 hold 27, 24, 16.5, 39.6 and 13.2. What does =COUNTIF(D2:D6,">20") return? [1]

A 3 B 2

C 5 D 90.6

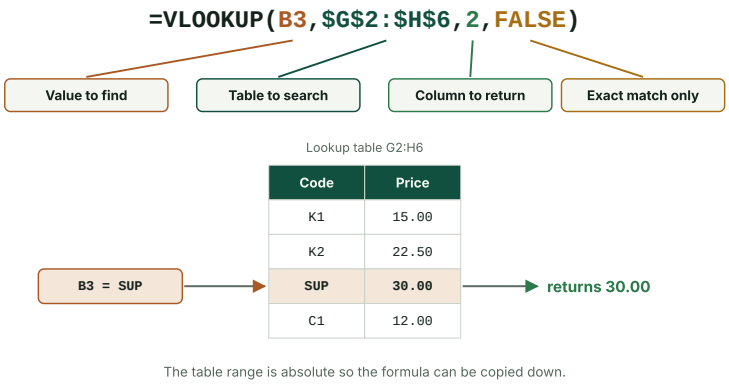
UNIT 1: DIGITAL TECHNOLOGY · SPREADSHEET APPLICATIONS

Spreadsheets 2: IF, VLOOKUP, validation, templates, modelling, charts and macros

L12 WORKSHEE

Name	Class	Date	Score

I can: Write IF, nested IF and VLOOKUP functions and explain what each part does · Use conditional formatting, data validation and templates, and explain why they help users · Use a spreadsheet model to answer what-if questions · Choose, create and label a suitable chart, and describe what a macro is and why it is useful



Use this diagram to help you: The four parts of a VLOOKUP formula and the row it matches in a lookup table

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **IF function**
- ☐ **Nested IF**
- ☐ **VLOOKUP**
- ☐ **Lookup table**
- ☐ **Conditional formatting**
- ☐ **Data validation**

A Vertical lookup: a function that searches for a value in the first column of a table and returns a value from a specified column in the same row. It updates automatically if the source data changes.

B A function that tests a condition and returns one value if it is true and another if it is false, e.g. =IF(A1>60,"PASS","FAIL").

C An IF function placed inside another IF function so that more than two outcomes are possible, e.g. =IF(B2>=80,"Distinction",IF(B2>=65,"Merit","Fail")).

D The range of cells searched by VLOOKUP; the values to be matched are in its first column.

E An automatic check when data is entered that it is sensible and within acceptable limits, e.g. a range check of 1 to 16. It cannot check the data is correct.

F Formatting (e.g. fill or font colour) that is applied automatically to cells only when a specified condition is met, e.g. hours worked over 40 highlighted.

B Fill the gaps

Use the word bank · 9 marks

IF | Conditional formatting | error alert | VLOOKUP | template | what-if | nested IF | validation | macro

The _____ function returns one value when a condition is true and another when it is false; placing one inside another makes a _____. _____ searches the first column of a lookup

table and returns a value from the same row. _____ changes a cell's appearance automatically when a rule is met. Data _____ limits what can be entered, and an _____ asks the user to try again. A _____ gives a ready-made layout for others to use. Changing inputs in a model to see the effect is called _____ analysis, and a _____ automates a repeated task.

C True or false?

Tick one box · 9 marks

	True	False
VLOOKUP searches the last column of the table.	☐	☐
What-if analysis is used in data modelling.	☐	☐
A chart should have axis titles.	☐	☐
In an IF function, the second part is what happens when the condition is false.	☐	☐
An IF statement placed inside another IF statement is said to be nested.	☐	☐
VLOOKUP can only return a value from the first column of the table.	☐	☐
Data validation can check that the data entered is correct.	☐	☐
With FALSE as the last part, VLOOKUP shows #N/A if the value is not in the table.	☐	☐
A macro can be created by recording the tasks to be carried out.	☐	☐

D Exam-style questions

72 marks

1 Cell D2 contains a test mark. Write an IF function that displays 'Merit' if the mark is 70 or more, otherwise 'Pass'. [2]

2 Explain what the function =IF(A1>60,"PASS","FAIL") does. State the result if A1 contains 60. [3]

3 Cell B2 contains a student's mark. Write a nested IF function for C2 that displays "Distinction" for a mark of 80 or more, "Merit" for a mark of 65 or more and "Fail" otherwise. [3]

4 An online shop charges £4.95 delivery, stored in K1, unless the order total in J2 is £50 or more, when delivery is free. Write an IF function for K2 that can be replicated down column K. State the result in K2 when J2 is £35.50. [3]

5 A Year 12 formal committee keeps a budget sheet. Cell C8 holds the amount allowed for the DJ and D8 the amount actually spent. Write an IF function for F8 that displays "Overspent" if the amount spent is more than the amount allowed, otherwise "OK". [2]

6 A travel club converts pounds to euros. H1 holds the base rate and H2 the bonus rate. Groups paying less than £100 (in column H, starting at H5) get the base rate; everyone else gets the bonus rate. Write the formula for I5 so that it can be replicated down column I, and explain why absolute references are needed. [4]

7 Explain what =VLOOKUP(B3,$F$2:$G$10,2,FALSE) does. [3]

8 A table of activities and costs is stored in A1:B5: Activity/Cost, Kayaking £12, Rock Climbing £10, Archery £8.50, Zip Line £15. Explain what =VLOOKUP("Rock Climbing",A1:B5,2,FALSE) does and state the value returned. [4]

9 Explain why the lookup table in a VLOOKUP formula is often written with absolute references, e.g. $A$2:$B$5. [2]

10 A leisure centre table in G20:I24 has Code in column G, Activity in column H and Price in column I. A booking code is entered in B20. (a) Write a VLOOKUP for C20 to show the price. (b) State what is displayed if a code is entered that is not in the table. [3]

11 Describe what is meant by conditional formatting and give an example of its use. [2]

12 Explain why conditional formatting is better than changing the colour of cells by hand. [2]

13 Values entered in B1:B6 must be between -£50 and +£100. Describe how data validation could be used and state one limitation of validation. [3]

14 In a Noughts and Crosses spreadsheet, state a suitable validation rule for the nine board cells. [2]

15 Describe one benefit of using a template. [2]

16 Describe how data could be imported into a spreadsheet from a database, and give one benefit of importing rather than typing the data. [3]

17 Explain what is meant by data modelling. Give an example of how a spreadsheet model could be used. [3]

18 A burger van owner has a spreadsheet model of income, costs and profit. Explain how the model could help the owner decide what price to charge. [4]

19 A sports camp model has fixed costs of £1,500 and costs of £18 per child. (a) Calculate the profit if 120 children book at £35 each. (b) A what-if test raises the price to £38 and 105 children book. Calculate the new profit. [4]

20 State three features a chart must have to be clearly labelled. [3]

21 State three types of chart and say what each is used to show. [3]

22 A youth club has its monthly attendance figures for January to June in cells B2 to B7, with the month names in A2 to A7. State the cell range that should be selected to create a chart, and name and justify the most suitable type of chart. [3]

23 What is a macro? Describe one benefit of using a macro in a spreadsheet. [3]

24 A Newry café uses a spreadsheet to take orders. The manager wants to make the sheet easier and more accurate for part-time staff to use. Evaluate how templates, data validation, VLOOKUP and macros could help. [6]

E Quick quiz

Circle one answer · 9 marks

1 =IF(B2>10,"Yes","No") shows what if B2 is 10? [1]

A Yes B No

C 10 D Error

2 In =VLOOKUP(B2,$G$2:$H$6,2,FALSE), what does the 2 tell the spreadsheet? [1]

A Which column of the table to search
B How many rows to check
C Which column of the table to return a value from
D Where to put the answer

3 Turning a cell red when stock is below 5 is... [1]

A conditional formatting
B a macro
C a VLOOKUP
D a template

4 Best chart to show temperature changes over a month? [1]

A Pie chart
B Scatter graph
C Line graph
D No chart

5 Peter scores 70. What does =IF(B2>=80,"Distinction",IF(B2>=65,"Merit","Fail")) show? [1]

A Distinction
B Merit
C Fail
D 70

6 Which Data Validation tab sets the message shown when invalid data is entered? [1]

A Settings
B Input Message
C Conditional Formatting
D Error Alert

7 Which file type must you use to save a workbook containing a macro? [1]

A .xlsm
B .xlsx
C .csv
D .xltx

8 Changing the price in a model to see the effect on profit is called... [1]

A validation
B importing
C conditional formatting
D what-if analysis

9 J2 holds £50.00. What does =IF(J2>=50,0,4.95) return? [1]

A 4.95
B 0
C 50
D FALSE

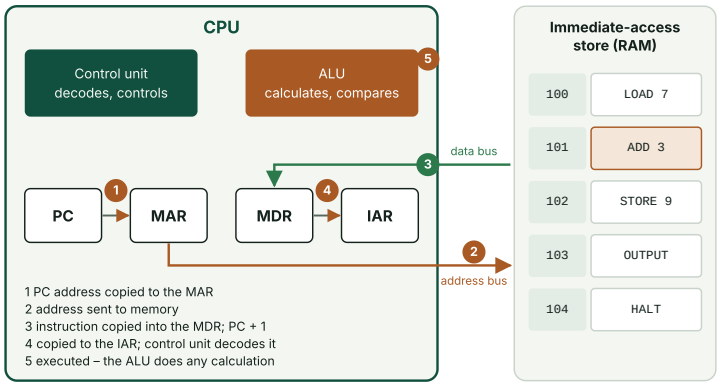
UNIT 1: DIGITAL TECHNOLOGY · COMPUTER HARDWARE

The CPU and the fetch–execute cycle

L13 WORKSHEET

Name	Class	Date	Score

I can: Explain the purpose of the CPU using processing language, not just 'the brain' · Describe what the control unit, ALU and immediate-access store each do · Describe the job of the PC, MAR, MDR, IAR and accumulator · Describe each stage of the fetch–decode–execute cycle in the correct order



Use this diagram to help you: Inside the CPU: the fetch–execute cycle using the PC, MAR, MDR and IAR with main memory

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **CPU**
- ☐ **ALU**
- ☐ **Control unit**
- ☐ **Immediate-access store**
- ☐ **Register**
- ☐ **Program counter**

A Main memory (RAM) holding the programs and data currently in use; each location has a unique address.

B Central processing unit. A microprocessor whose purpose is to process data and carry out the instructions of programs, managing the functions of the computer.

C Arithmetic logic unit: the part of the CPU that carries out arithmetic calculations and logical operations and comparisons.

D The part of the CPU that fetches and decodes instructions and sends control signals to co-ordinate the other components.

E Register holding the address of the next instruction to be fetched; incremented after each fetch.

F A very small, very fast memory location inside the CPU used for a specific purpose.

B Fill the gaps

Use the word bank · 8 marks

MAR | microprocessor | accumulator | MDR | ALU | program counter | control unit | decode

The CPU is a _____ that processes data and carries out program instructions. The _____ decodes each instruction and sends control signals. The _____ carries out calculations and logical comparisons, and stores each result in the _____. The _____ holds the address of

the next instruction. The _____ holds the address being accessed, while the _____ holds the data or instruction fetched from memory. The cycle is fetch, _____, execute.

C True or false?

Tick one box · 9 marks

	True	False
The ALU decodes instructions.	☐	☐
The program counter is increased by one after each instruction is fetched.	☐	☐
The immediate-access store holds the programs and data currently in use.	☐	☐
The MAR holds the data that has just been fetched from memory.	☐	☐
The CPU is a microprocessor.	☐	☐
Writing 'the CPU is the brain of the computer' is enough to earn the marks for its purpose.	☐	☐
The result of an ALU calculation is first stored in the accumulator.	☐	☐
Every location in main memory has its own unique address.	☐	☐
The stages of the cycle happen in the order fetch, execute, decode.	☐	☐

D Exam-style questions

33 marks

1 Explain the purpose of the central processing unit (CPU). [2]

2 State the three main components of the CPU. [3]

3 Describe the role of the arithmetic logic unit (ALU). Give an example of an operation it carries out. [3]

4 Describe the role of the control unit. [3]

5 What is the immediate-access store? Explain why it is needed for the fetch–execute cycle. [2]

6 Expand the acronyms MAR and MDR, and explain the difference between them. [4]

7 State what is held in the program counter, and describe what happens to it after an instruction is fetched. [2]

8 Describe the fetch stage of the fetch–execute cycle, referring to the program counter, MAR, MDR and IAR. [4]

9 A café in Enniskillen uses a till that adds up the price of each order. Describe what happens in the decode and execute stages when the till's CPU runs an instruction to add the price of a drink to the running total. [4]

10 Discuss how the components of the CPU and its registers work together to carry out a program. Your answer should refer to the control unit, the ALU, the immediate-access store and the fetch–execute cycle. [6]

E Quick quiz

Circle one answer · 8 marks

1 Which part of the CPU carries out calculations and logical comparisons? [1]

A Control unit

B Arithmetic logic unit (ALU)

C Memory address register
D Program counter

2 Which register holds the address of the next instruction to be fetched? [1]

A Program counter
B Memory data register
C Accumulator
D Instruction address register

3 What does the memory data register (MDR) hold? [1]

A The address of the location being accessed
B The address of the next instruction
C The instruction or data just fetched from, or about to be written to, memory
D The clock speed of the CPU

4 Which component decodes instructions and sends control signals? [1]

A ALU
B Accumulator
C Immediate-access store
D Control unit

5 What is the correct order of the cycle? [1]

A Execute, fetch, decode
B Fetch, decode, execute
C Decode, fetch, execute
D Fetch, execute, decode

6 Which best explains the purpose of the CPU? [1]

A It stores files permanently
B It displays output to the user
C It processes data and carries out the instructions of programs
D It connects the computer to a network

7 In the first step of the fetch stage, the contents of the program counter are copied into... [1]

A the MAR
B the MDR
C the accumulator
D the ALU

8 Where is the result of an ALU calculation first held? [1]

A Program counter
B Memory address register
C Hard disc
D Accumulator

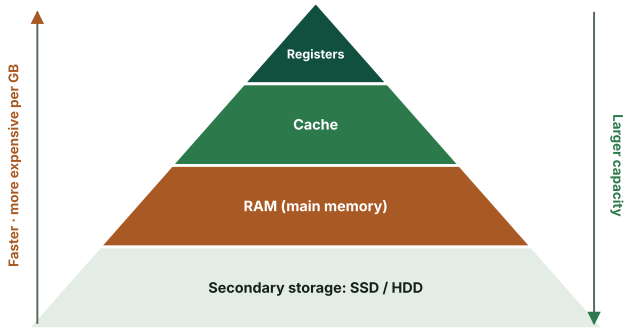
UNIT 1: DIGITAL TECHNOLOGY · COMPUTER HARDWARE

CPU performance and memory: RAM, ROM and cache

L14 WORKSHEET

Name	Class	Date	Score

I can: Explain how clock speed, cache size and number of cores each affect CPU performance · Explain the purpose of RAM, ROM and cache · Compare RAM, ROM and cache using the terms volatile and non-volatile · Recommend and justify a processor and memory specification for a given user



ROM is separate: small, non-volatile, holds the boot-up instructions.

Use this diagram to help you: Memory hierarchy pyramid from registers to secondary storage

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Clock speed**
- ☐ **Core**
- ☐ **Cache**
- ☐ **RAM**
- ☐ **ROM**
- ☐ **Volatile**

A Small, very fast memory on or near the CPU that holds frequently used instructions and data, so they need not be fetched from slower RAM.

B Random access memory: volatile read/write main memory that holds the operating system, programs and data currently in use.

C An independent processing unit inside a CPU; each core has a control unit, an ALU and registers, so it can process instructions at the same time as other cores.

D Describes memory that loses its contents when the power is switched off, such as RAM and cache.

E The number of fetch–execute cycles a CPU can carry out per second, measured in hertz (e.g. 3 GHz = 3 billion cycles per second).

F Read only memory: non-volatile memory whose contents cannot normally be changed, used to store the boot-up instructions (BIOS).

B Fill the gaps

Use the word bank · 8 marks

RAM | boot | volatile | cores | hertz | Cache | ROM | virtual memory

Clock speed is measured in _____, and a 3 GHz processor carries out three billion cycles per second. A CPU with more _____ can process several instructions at the same time. _____ is a small amount of very fast memory that holds frequently used instructions and data. _____

holds the programs and data currently in use and is _____, so its contents are lost when the power is off. _____ is non-volatile and stores the _____ program. When RAM is full, the operating system uses _____ on the storage drive.

C True or false?

Tick one box · 9 marks

	True	False
ROM is volatile.	☐	☐
A larger cache usually improves performance.	☐	☐
RAM loses its contents when the power is switched off.	☐	☐
Clock speed is the number of cores in a processor.	☐	☐
A 2 GHz processor can carry out two billion cycles per second.	☐	☐
Doubling the number of cores always doubles the speed of every program.	☐	☐
ROM holds the instructions needed to start up the computer.	☐	☐
Cache is slower than RAM but has a larger capacity.	☐	☐
Using virtual memory slows a computer down because storage is slower than RAM.	☐	☐

D Exam-style questions

28 marks

1 State the unit used to measure clock speed. [1]

2 Describe the impact of increasing the clock speed on the performance of a CPU. [2]

3 A computer in the school library has a 3.2 GHz processor. State how many cycles per second this is, written in full. [1]

4 Explain how having more cores can improve performance, and give one reason why doubling the number of cores might not double the speed. [3]

5 Two otherwise identical laptops in a Newry shop have 4 MB and 12 MB of cache. Explain what cache memory is for and why the 12 MB laptop is likely to perform better. [3]

6 Explain the purpose of RAM. [2]

7 Explain the purpose of ROM. Include its role when the computer is switched on. [3]

8 Describe two differences between RAM and ROM. [4]

9 A pupil's laptop slows down when many programs are open. Explain why adding more RAM could help. [3]

10 A sports club in Omagh wants a new computer to edit match videos for its website. Compare two possible specifications: Computer A (2.6 GHz, 4 cores, 6 MB cache, 8 GB RAM) and Computer B (3.4 GHz, 8 cores, 16 MB cache, 16 GB RAM). Discuss which is more suitable, referring to clock speed, cores, cache and RAM. [6]

E Quick quiz

Circle one answer · 8 marks

1 Clock speed is measured in... [1]

A bytes
B hertz
C pixels
D bits per second

2 Why does a larger cache usually improve performance? [1]

A It increases the clock speed
B It adds more cores

C More frequently used data is kept close to the CPU, so there are fewer slow trips to RAM

D It stores the operating system permanently

3 Which type of memory is volatile and holds the programs currently in use? [1]

A ROM

B SSD

C Blu-ray

D RAM

4 What is stored in ROM? [1]

A The start-up (boot) instructions

B Your saved documents

C The web pages you are viewing

D Frequently used data copied from RAM

5 Extra cores help most when... [1]

A the screen resolution is higher

B software can split its work to run in parallel

C the computer is switched off

D the ROM is larger

6 A processor runs at 4 GHz. How many cycles can it carry out per second? [1]

A 4,000

B 4,000,000

C 4,000,000,000

D 400,000,000

7 Which statement about cache is correct? [1]

A It is faster than RAM and has a smaller capacity

B It is non-volatile

C It is part of the hard disc

D It is read only

8 What is virtual memory? [1]

A Memory inside the CPU

B Another name for ROM

C Memory on a cloud server

D Part of the storage drive used as extra RAM when RAM is full

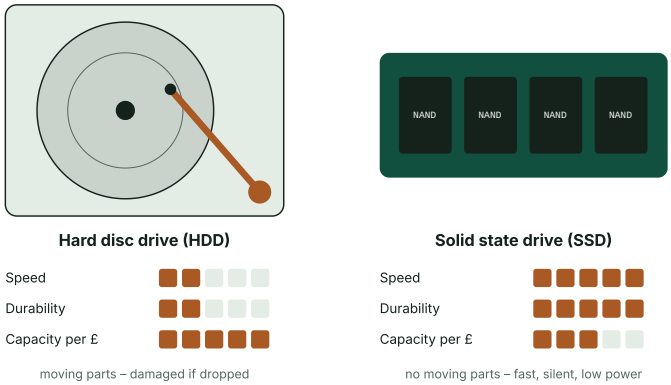
UNIT 1: DIGITAL TECHNOLOGY · COMPUTER HARDWARE

Input, output and storage devices

L15 WORKSHEET

Name	Class	Date	Score

I can: Describe how each device in the specification works and what it is used for · Give advantages and disadvantages of each device, linked to a context · Compare HDD, SSD and HD optical storage using capacity, speed, durability, portability and cost · Recommend devices for a scenario and justify the choice



Use this diagram to help you: Hard disc drive with spinning platter compared with a solid state drive made of memory chips

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Input device**
- ☐ **Output device**
- ☐ **Graphics digitiser**
- ☐ **Touch screen**
- ☐ **Laser printer**
- ☐ **3D printer**

A A display with a touch-sensitive surface that works as both an input and an output device.

B An output device that builds a solid object layer by layer from a computer-aided design (CAD) model.

C Hardware used to present information from a computer, such as speakers or a printer.

D An input device made of a flat pad and a stylus, used to capture freehand drawings or signatures.

E Hardware used to enter data into a computer, such as a microphone, mouse or graphics digitiser.

F An output device that uses a laser to form the page on a charged drum, which attracts toner that is fused onto paper; fast and high quality.

B Fill the gaps

Use the word bank · 7 marks

touch screen | 3D printer | SSD | HDD | graphics digitiser | laser printer | Blu-ray

A _____ is used with a stylus to draw freehand. A _____ is both an input and an output device. A _____ uses toner and a drum to produce high-quality pages quickly. A _____ builds solid objects layer by layer from a CAD design. An _____ uses flash memory and has no moving parts, while an _____ stores data magnetically and has a lower cost per gigabyte. _____ discs use a blue-violet laser to store high-definition films.

C True or false?

Tick one box · 9 marks

	True	False
SSDs are more durable than HDDs because they have no moving parts.	☐	☐
A laser printer is an input device.	☐	☐
Blu-ray is a type of high-definition optical storage media.	☐	☐
3D printers are fast and economical for making large numbers of large objects.	☐	☐
Voice recognition accuracy can drop when there is background noise.	☐	☐
An HDD usually has a lower cost per gigabyte than an SSD.	☐	☐
A graphics digitiser is better than a mouse for selecting items from menus.	☐	☐
Laser printers can print onto multipart carbon-copy forms.	☐	☐
A Blu-ray disc can store more than a DVD because its laser makes smaller pits that are packed more closely.	☐	☐

D Exam-style questions

31 marks

1 State whether each of the following is an input device, an output device or both: microphone, speakers, touch screen. [3]

2 Describe one advantage and one disadvantage of using a touch screen to take orders in a fast-food restaurant. [2]

3 Describe what a graphics digitiser is and give a typical use. [2]

4 Aoife, a GCSE pupil in a busy family home in Lurgan, plans to dictate her homework into a microphone using voice recognition instead of typing it. Describe two problems she might have. [4]

5 Give two advantages of a laser printer for a school office that prints large numbers of letters to parents. [2]

6 The Royal Victoria Hospital in Belfast is thinking of buying a 3D printer. Explain how a 3D printer makes an item and suggest one way the hospital could use it. [3]

7 Explain why an SSD is more suitable than an HDD for a laptop used by a building-site inspector. [2]

8 Give one advantage of an HDD over an SSD. [1]

9 Apart from an SSD, flash memory is found in other everyday devices. Identify two of these, describe each one and give a situation in which a teenager might use it. [4]

10 Explain why a Blu-ray disc can store more data than a DVD of the same size. [2]

11 A council leisure centre in Derry~Londonderry is setting up a reception desk. Staff will book classes, print membership cards and store records of 5,000 members. Discuss suitable input, output and storage devices for the desk, justifying each choice. [6]

E Quick quiz

Circle one answer · 8 marks

1 Which storage device has no moving parts? [1]

A Hard disc drive
B Solid state drive
C Blu-ray drive
D DVD drive

2 How does a 3D printer create an object? [1]

A It prints toner onto paper using a drum
B It scans a real object with a laser
C It builds the object one layer at a time
D It carves the object from a solid block

3 Which device is both an input and an output device? [1]

A Touch screen
B Microphone
C Speakers
D Laser printer

4 Blu-ray is an example of... [1]

A magnetic storage
B solid state storage
C RAM
D high-definition optical storage

5 What is the main advantage of an HDD over an SSD? [1]

A Faster access speed
B More durable
C Lower cost per gigabyte
D Silent operation

6 Which device would best suit a designer who needs to draw detailed artwork by hand? [1]

A Graphics digitiser
B Microphone
C Mouse
D Speakers

7 Which is a disadvantage of using a microphone for voice input? [1]

A It is slower than typing
B Background noise can cause words to be recognised wrongly
C It needs a flat surface
D It cannot be used hands-free

8 Why is a laser printer a good choice for a busy school office? [1]

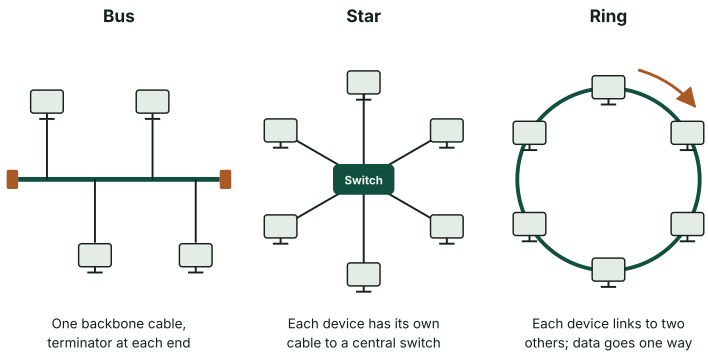
A It can print 3D objects
B It is cheap to buy in colour
C It can print carbon-copy forms
D It prints large volumes quickly at high quality

Networks: LAN, WAN, internet, hardware and topologies

L16 WORKSHEET

Name	Class	Date	Score

I can: Compare a LAN and a WAN and give examples of each · Weigh up the advantages and disadvantages of a network for an organisation · Explain how the internet, the World Wide Web, an intranet and the Internet of Things differ · Describe what a NIC, network cables, a switch and a router do · Describe and compare the bus, star and ring topologies



Use this diagram to help you: Bus, star and ring network topologies

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **LAN**
- ☐ **WAN**
- ☐ **Standalone computer**
- ☐ **Internet**
- ☐ **World Wide Web**
- ☐ **Intranet**

A The global network of networks: the hardware and connections that carry data between computers worldwide.

B A computer that is not connected to a network, so its files and devices cannot be shared.

C A private network owned by an organisation that uses internet technology but can only be accessed by authorised users who log on.

D Wide area network: a network covering a large geographical area that uses telecommunications links such as fibre optic cables or satellites.

E Local area network: a network covering a small geographical area such as one building or site, usually owned by the organisation.

F A collection of linked web pages stored on web servers and accessed over the internet using a browser.

B Fill the gaps

Use the word bank · 8 marks

switch | NIC | star | terminator | WAN | router | LAN | intranet

A _____ covers a small area such as one school, while a _____ covers a large geographical area using telecommunications links. Every computer needs a _____ to connect to the network. Within the LAN a _____ forwards data to the intended computer, and a _____

connects the LAN to the internet using IP addresses. In a bus network a _____ at each end of the backbone stops signals bouncing back. In a _____ network every computer has its own cable to a central device. A private network that only authorised staff can log on to is an _____.

C True or false?

Tick one box · 8 marks

	True	False
The World Wide Web and the internet are the same thing.	☐	☐
IoT stands for Internet of Things.	☐	☐
In a ring network, one broken cable can stop the whole network.	☐	☐
A switch connects a LAN to the internet.	☐	☐
A NIC lets a computer connect to a network.	☐	☐
In a bus network, if the backbone cable fails only one computer is affected.	☐	☐
Anyone with an internet connection can use a company's intranet.	☐	☐
A WAN usually uses telecommunications links such as fibre optic cables or satellites.	☐	☐

D Exam-style questions

36 marks

1 Southern Regional College links the computers in each of its campuses, and also links its Newry, Armagh and Banbridge campuses together. Describe two differences between the network inside one campus (a LAN) and the network joining the campuses (a WAN). [4]

2 Explain the difference between the internet and the World Wide Web. [2]

3 A grammar school in Coleraine has an intranet for its pupils and staff. Explain how the school's intranet differs from the internet. [4]

4 Expand the abbreviation IoT and describe two ways a teenager might use IoT devices at home. [5]

5 Describe the function of a network interface card (NIC). [2]

6 A GP surgery in Banbridge has a LAN connected to the internet. Describe the function of the switch and the router in this network. [4]

7 A small office in Cookstown has four PCs connected in a bus network. Describe how this network is laid out and explain why it needs terminators. [4]

8 Describe a star topology and give one advantage and one disadvantage of it. [3]

9 Explain why adding a new computer to a ring network is more difficult than adding one to a star network. [2]

10 A garden centre in Lisburn has six standalone computers and two printers. The owner is thinking about replacing them with a network. Discuss the advantages and disadvantages of doing this. [6]

E Quick quiz

Circle one answer · 9 marks

1 Which device connects a school's LAN to the internet and uses IP addresses to send data along the best route? [1]

A Switch
B Router
C Network interface card
D Terminator

2 In which topology is every computer connected by its own cable to a central switch? [1]

A Bus
B Ring
C Star
D Backbone

3 What is the job of a terminator in a bus network? [1]

A To connect the bus to the internet
B To stop signals bouncing back along the backbone
C To speed up the backbone
D To give each computer an IP address

4 What does IoT stand for? [1]

A Internet of Technology
B Intranet of Things
C Internet of Things
D Input of Things

5 Which best describes the World Wide Web? [1]

A The cables and routers linking computers worldwide
B A collection of linked web pages accessed over the internet
C A private network for one company
D A type of LAN

6 Which is a disadvantage of a network compared with standalone computers? [1]

A Files can be backed up centrally
B Printers can be shared
C A virus can spread quickly to all computers
D Software can be installed once for everyone

7 Which statement about a ring network is true? [1]

A Data travels in one direction around the loop
B It needs a terminator at each end
C Each computer connects to a central switch
D A fault in one cable affects only one computer

8 A WAN differs from a LAN because a WAN... [1]

A is always wireless
B covers a large geographical area using telecommunications links
C cannot connect to the internet
D only links two computers

9 Which device inside a computer provides the port for a network cable? [1]

A Router
B Switch
C File server
D Network interface card

UNIT 1: DIGITAL TECHNOLOGY · NETWORK TECHNOLOGIES

Network communication technologies

L17 WORKSHEET

Name	Class	Date	Score

I can: Describe how Wi-Fi, Bluetooth, optical fibre, 4G and 5G transmit data · Give advantages, disadvantages and typical uses of each technology · Compare wireless and optical fibre for cost, speed and reliability · Choose and justify the best technology for a scenario



Use this diagram to help you: Typical range of Bluetooth, Wi-Fi, 4G, 5G and optical fibre

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Wi-Fi**
- ☐ **Bluetooth**
- ☐ **Optical fibre**
- ☐ **4G**
- ☐ **5G**
- ☐ **Bandwidth**

A The amount of data that can be transmitted in a given time, measured in bits per second.

B Cable made of thin glass strands that carries data as pulses of light, giving very high bandwidth and immunity to electrical interference.

C Wireless technology that uses radio waves and a wireless router or access point to connect devices to a network without cables.

D Fourth-generation mobile technology giving high-speed wireless data through a network of masts.

E Short-range, low-power wireless technology that connects two nearby devices directly after they have been paired.

F Fifth-generation mobile technology with faster speeds, much lower latency and capacity for more devices than 4G, but needing more masts.

B Fill the gaps

Use the word bank · 8 marks

interference · repeater · latency · Bluetooth · Wi-Fi · Optical fibre · 5G · bandwidth

_____ uses radio waves and a wireless router so devices can join a network without cables, but the signal weakens with distance. _____ connects two nearby devices directly once they have been paired. _____ sends data as pulses of light along glass strands, so it is not affected by electrical _____. Over very long distances a _____ boosts the signal. _____ has

much lower _____ than 4G, meaning less delay. The amount of data that can be sent in a given time is called _____.

C True or false?

Tick one box · 8 marks

	True	False
Optical fibre is affected by electrical interference.	☐	☐
Bluetooth is suitable for connecting wireless earbuds to a phone.	☐	☐
Bluetooth needs a router before two devices can communicate.	☐	☐
The further a device is from a wireless router, the weaker the Wi-Fi signal.	☐	☐
Wi-Fi is always more secure than a cabled connection.	☐	☐
5G has lower latency than 4G.	☐	☐
Each high-speed 5G mast covers a larger area than a 4G mast.	☐	☐
Optical fibre is expensive to install.	☐	☐

D Exam-style questions

40 marks

1 Describe Bluetooth technology and give one typical use. [3]

2 A family in Larne connect their phones, tablets and smart TV to the internet using Wi-Fi. State two features of the wireless technology they are using. [2]

3 Give two advantages and one disadvantage of optical fibre as a communications medium. [3]

4 Explain one advantage of 5G over 4G. [2]

5 A café in Portrush wants to offer free internet access to customers. Explain why Wi-Fi is suitable and describe one security measure the owner should take. [4]

6 Explain why optical fibre is a better choice than Wi-Fi for linking the two main buildings of a hospital in Derry~Londonderry. [4]

7 A farmer near Enniskillen wants to monitor animals in remote fields using IoT sensors. Explain why 4G or 5G may be more suitable than Wi-Fi, and describe one problem the farmer could face. [4]

8 A secondary school in Armagh is choosing between Wi-Fi and optical fibre cabling to connect its new sixth-form centre to the main building. Compare the two options, referring to cost, speed and reliability. [6]

9 Evaluate the use of Wi-Fi in a secondary school. [4]

10 A Belfast events company streams live concerts and needs staff to share video and data at different venues. Evaluate the effectiveness of Wi-Fi, optical fibre and 5G for the company. [8]

E Quick quiz

Circle one answer · 9 marks

1 A hospital needs a very fast link between two buildings that cannot be affected by electrical interference from medical equipment. Which technology is best? [1]

A Bluetooth
B Optical fibre
C Wi-Fi
D 4G

2 A pupil wants to connect wireless earbuds to their phone. Which technology will they use? [1]

A Bluetooth
B Optical fibre
C 5G
D A network cable

3 What is a key benefit of 5G compared with 4G? [1]

A Each mast covers a larger area
B Much lower latency
C No masts are needed
D It works without a SIM or network

4 What is latency? [1]

A The amount of data sent per second
B The range of a signal
C The delay before data arrives
D The number of users on a network

5 Which is a disadvantage of Wi-Fi? [1]

A It needs a cable to every device
B It cannot connect phones
C It only works for headphones
D Signal is weakened by walls and distance

6 Why is optical fibre considered secure? [1]

A It is very hard to tap into the cable
B It is password protected
C It only works over 10 metres
D It uses radio waves

7 Before two Bluetooth devices can share data they must be... [1]

A connected to a router
B plugged into a switch
C paired
D given a URL

8 Which unit is used to measure Wi-Fi speed? [1]

A Megabytes of storage (MB)
B Gigahertz (GHz)
C Pixels per inch
D Megabits per second (Mbps)

9 A farmer in a remote field with no Wi-Fi needs internet access for sensors. What is the most likely choice? [1]

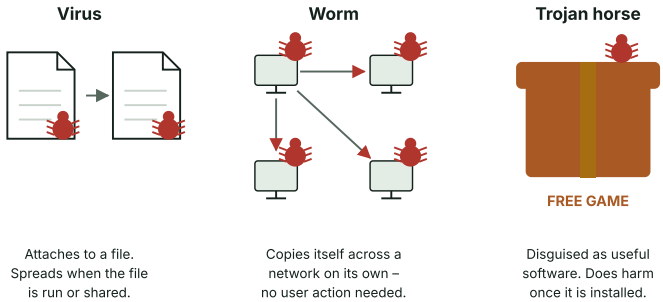
A Bluetooth
B Mobile data (4G/5G)
C Optical fibre to each sensor
D A bus network

Cybercrime and malware

L18 WORKSHEET

Name	Class	Date	Score

I can: Define cybercrime and describe each of the nine threats to cybersecurity in the specification · Recognise the threat being described in a scenario and explain its effect on a person or organisation · Define malware and describe how it gets onto a computer · Distinguish between a virus, worm, Trojan horse, key logger and spyware · Explain how anti-virus software protects a computer



Also: key loggers record keystrokes · spyware secretly sends your data to others.

Use this diagram to help you: How a virus, a worm and a Trojan horse spread

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Cybercrime**
- ☐ **Hacking**
- ☐ **Phishing**
- ☐ **Denial of service**
- ☐ **Cyber stalking**
- ☐ **Cyber defamation**

A Posting untrue statements online that harm the good name of a person or an organisation.

B An attack that floods a server or website with requests so that it cannot respond to genuine users.

C Any crime committed using a computer, a network or the internet.

D Emails, texts or websites that pretend to come from a trusted organisation in order to trick people into giving personal details such as passwords or bank details.

E Gaining unauthorised access to a computer system or its data, for example to steal, change or delete data.

F Using technology to repeatedly harass, track or frighten a person.

B Fill the gaps

Use the word bank · 8 marks

worm | virus | Trojan horse | hacking | denial of service | cybercrime | phishing | key logger

Using a computer or the internet to commit a crime is called _____. Gaining unauthorised access to a system is _____. Fake emails pretending to be from a bank, asking you to confirm your details, are an example of _____. Flooding a server with requests so genuine users cannot get in is a _____ attack. A _____ copies itself and spreads across a network without needing a

host file, while a _____ attaches itself to a file. A _____ is disguised as useful software, and a _____ secretly records every key pressed.

C True or false?

Tick one box · 8 marks

	True	False
A virus needs a host file to spread.	☐	☐
A worm needs the user to open an infected file before it can spread.	☐	☐
Cyber defamation means publishing false, damaging statements about someone online.	☐	☐
A denial of service attack always steals data from the server.	☐	☐
A Trojan horse copies itself to other computers automatically.	☐	☐
Spyware can slow a computer down because it uses memory and processing power.	☐	☐
Spam is bulk unwanted email, often adverts or scams.	☐	☐
Anti-virus software only needs to be installed once and never updated.	☐	☐

D Exam-style questions

29 marks

1 Define the term cybercrime. [1]

2 Describe phishing. [2]

3 A Newry sports shop sells kit through its website. Describe how a denial of service attack would affect the shop. [2]

4 Define malware. [2]

5 Describe the difference between a virus and a worm. [2]

6 Explain how a key logger could be used to steal money from a bank account. [2]

7 Describe two ways that malware could get onto a pupil's laptop. [2]

8 Describe the difference between a Trojan horse and spyware. [4]

9 A youth club in Derry~Londonderry has a social media page and stores members' details on a laptop. Name and describe two threats to cybersecurity, other than malware, that the club could face. [4]

10 Explain why anti-virus software must be updated regularly. [2]

11 A family-run gift shop in Enniskillen has recently started selling online. Discuss the threats that phishing and denial of service attacks pose to the shop and its customers. [6]

E Quick quiz

Circle one answer · 8 marks

1 Malware that is disguised as a useful program, so the user installs it themselves, is a... [1]

A worm
B Trojan horse
C spam
D key logger

2 A text claiming to be from your bank asks you to click a link and 'verify' your PIN. What is this? [1]

A Spamming
B Hacking
C Phishing
D Cyber defamation

3 Which malware spreads across a network by itself without a host file? [1]

A Worm
B Virus

C Trojan horse
D Spyware

4 Flooding a website with requests so genuine customers cannot use it is... [1]

A digital forgery
B cyber stalking
C data theft
D denial of service

5 Publishing false, damaging claims about a person or business online is... [1]

A cyber defamation
B spamming
C phishing
D hacking

6 Which malware records every key a user presses? [1]

A Worm
B Virus
C Key logger
D Firewall

7 Stolen customer details used to apply for a credit card in someone else's name is... [1]

A identity theft
B spamming
C a worm
D digital forgery

8 Editing a scanned passport photo to make a fake document is... [1]

A cyber stalking
B digital forgery
C denial of service
D spyware

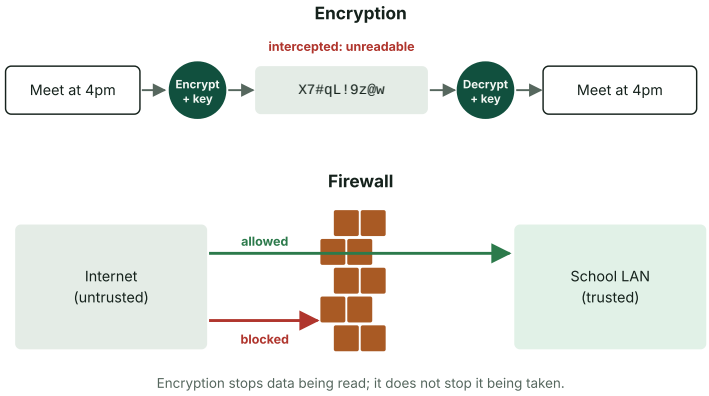
UNIT 1: DIGITAL TECHNOLOGY · CYBERSPACE, NETWORK SECURITY AND DATA TRANSFER

Protecting networks and data; protocols

L19 WORKSHEET

Name	Class	Date	Score

I can: Explain how encryption protects data that is intercepted or stolen · Describe how usernames, strong passwords and levels of access stop unauthorised use of a network · Explain how backups and firewalls protect a network and its data · Describe what a protocol is and why devices need one · Describe the purpose of FTP, HTTP and HTTPS and compare HTTP with HTTPS



Encryption stops data being read; it does not stop it being taken.

Use this diagram to help you: Encryption makes intercepted data unreadable; a firewall filters network traffic

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Encryption**
- ☐ **Decryption**
- ☐ **Firewall**
- ☐ **Access rights / levels of access**
- ☐ **Read only**
- ☐ **Strong password**

A Turning encrypted data back into readable form using the correct key.

B Permissions linked to a user ID that control what each user can see and do, e.g. no access, read only, read and write.

C A long password that mixes upper and lower case letters, digits and symbols and is not a dictionary word or personal detail.

D Scrambling data using an algorithm and a key so that it cannot be understood by anyone who does not have the key.

E Hardware or software that monitors and filters data entering or leaving a network, blocking traffic that does not meet security rules.

F An access level where a user can view data but cannot change it.

B Fill the gaps

Use the word bank · 10 marks

protocol | firewall | read only | FTP | Levels of access | strong password | two-factor authentication | backup | Encryption | HTTPS

_____ scrambles data so that it cannot be understood without the key. Each user logs on with a unique user ID and a _____, and _____ adds a second check such as a code sent to a

phone. _____ control what each user can do, for example a pupil may have _____ access to a file. A _____ is a copy of data that can be restored if the original is lost. A _____ filters data entering or leaving a network. A _____ is an agreed set of rules for transmitting data. _____ is used to transfer files, and _____ encrypts web traffic.

C True or false?

Tick one box · 9 marks

	True	False
HTTPS is less secure than HTTP.	☐	☐
Levels of access limit what each user can do with the data.	☐	☐
Encryption stops data being intercepted.	☐	☐
Two-factor authentication needs two different forms of verification.	☐	☐
A firewall scans files on the hard drive and removes viruses.	☐	☐
A backup should be stored in a different location from the original data.	☐	☐
FTP is used to transfer files between computers over a network.	☐	☐
A protocol is a type of network cable.	☐	☐
A user with read-only access can view a file but cannot change it.	☐	☐

D Exam-style questions

34 marks

1 Explain how encryption protects data. [2]

2 Describe the purpose of a firewall. [2]

3 Explain how levels of access protect patient data in a hospital. [2]

4 Describe the role of a protocol in data transfer. [2]

5 Expand FTP and describe its purpose. [2]

6 Describe the difference between HTTP and HTTPS. [2]

7 State three features of a strong password. [3]

8 Identify and describe three different levels of access a school network could give to its users. [6]

9 A dental surgery in Omagh stores patient records on its network server. Describe how backups should be carried out so that the records can be recovered if the server fails. [3]

10 Describe two measures, other than passwords, that a network logon system can use to reduce the risk of unauthorised access. [2]

11 Explain why an online shop should use HTTPS on its checkout page. [2]

12 A leisure centre run by a local council keeps members' personal and payment details on its network and lets members book classes online. Discuss how the council could use encryption, passwords, levels of access, backup and firewalls to protect this data. [6]

E Quick quiz

Circle one answer · 8 marks

1 What does a firewall do? [1]

A Removes viruses from infected files

B Monitors and filters network traffic, blocking anything that breaks the security rules

C Makes a copy of data each night

D Encrypts all passwords

2 How does HTTPS differ from HTTP? [1]

A HTTPS encrypts the data sent

B HTTPS is only used for files

C HTTPS needs no web browser

D HTTPS works without the internet

3 Which protocol is used to upload a website's files to a web server? [1]

A HTTP

B HTTPS

C FTP

D TCP

4 A protocol is... [1]

A a type of malware

B a network cable

C a strong password

D an agreed set of rules for transmitting data

5 Which is the strongest password? [1]

A password123

B Liam2010

C t7#Rq!9vLm2p

D fermanagh

6 A pupil can open the class notes but cannot change them. What access level is this? [1]

A Read only

B Read and write

C Full control

D No access

7 Why should a backup be stored off-site or in the cloud? [1]

A It makes the backup run faster

B A fire or theft at the building will not destroy both copies

C It encrypts the data

D It stops hackers logging in

8 What happens to encrypted data if it is intercepted? [1]

A It is deleted automatically

B It is sent back to the sender

C It cannot be understood without the key

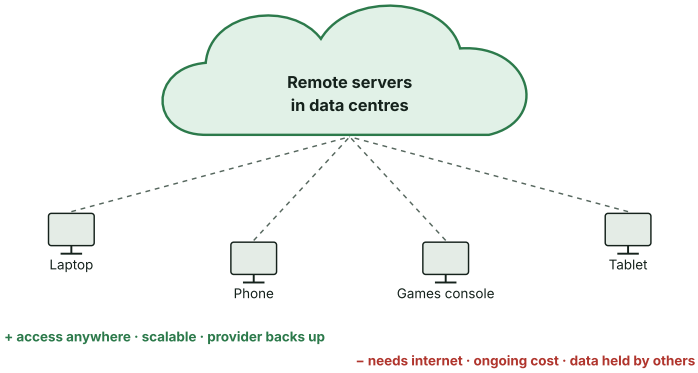
D It becomes a virus

Cloud technology

L20 WORKSHEET

Name	Class	Date	Score

I can: Define cloud computing and give examples of cloud services · Describe the advantages and disadvantages of the cloud for an organisation · Evaluate the impact of the cloud on gaming for players and game producers · Explain how cloud storage and collaborative tools change the way people store and share files



Use this diagram to help you: Cloud computing: devices reach remote servers over the internet

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Cloud computing**
- ☐ **Cloud storage**
- ☐ **Data centre**
- ☐ **Scalability**
- ☐ **Pay as you go**
- ☐ **Downtime**

A A building containing large numbers of servers that store data and run cloud services.

B Storing data and running software on remote servers that are accessed on demand over the internet, instead of on a local computer.

C The ability to increase or decrease storage, bandwidth or users quickly as an organisation's needs change.

D A period when a service is unavailable, for example because of an internet failure or provider maintenance.

E Keeping files on a provider's remote servers so they can be accessed from any device with an internet connection.

F A payment model where an organisation pays only for the cloud resources it actually uses.

B Fill the gaps

Use the word bank · 8 marks

downtime · internet · collaboration · Synchronisation · scalable · cloud gaming · version history · Cloud computing

_____ means using storage and software on remote servers accessed over the internet. Because storage can be increased or reduced as needed, the cloud is _____. Its main weakness is that it needs an _____ connection; a period when the service cannot be used is called _____. In _____, the game runs on the provider's servers and is streamed to the player's device.

_____ makes sure the latest version of a file appears on every device, and _____ lets users restore an earlier copy. Several users editing a document at the same time is an example of _____.

C True or false?

Tick one box · 8 marks

	True	False
Cloud services can be used without any internet connection.	☐	☐
Cloud gaming lets high-quality games run on low-powered devices.	☐	☐
Cloud providers usually take responsibility for backing up data.	☐	☐
Downtime is a risk of using cloud services.	☐	☐
Data stored in the cloud is guaranteed to be completely secure.	☐	☐
Synchronisation means a change made on one device appears on the user's other devices.	☐	☐
An organisation using the cloud no longer has to follow data protection law.	☐	☐
Selling games through the cloud can reduce piracy compared with selling discs.	☐	☐

D Exam-style questions

39 marks

1 Define the term cloud computing. [2]

2 A small accountancy firm in Newry keeps all its files on one ageing server in the office. State three benefits the firm would gain by switching to cloud computing. [3]

3 A solicitor's office in Armagh is moving its files to the cloud. Describe two disadvantages of cloud computing for the office. [4]

4 Explain what is meant by 'downtime' and why it is a risk for a business that relies on the cloud. [2]

5 Sinéad saves her GCSE coursework to OneDrive instead of a USB stick. Describe how her files are stored in the cloud and explain why she is less likely to lose her work. [4]

6 Describe one impact of cloud computing on file sharing. [2]

7 Explain how synchronisation and version history help a group of pupils working on a shared presentation. [4]

8 Explain why cloud gaming may not be suitable for someone living in a rural area. [2]

9 Describe two ways in which cloud computing benefits game producers. [4]

10 Conor, a Year 11 pupil in Strabane, is deciding whether to sign up to a cloud gaming service instead of buying a new console. Evaluate the impact cloud gaming would have on him as a player. [6]

11 A charity has volunteers in Belfast, Ballymena and Enniskillen working on a fundraising project. Discuss the benefits and drawbacks of using cloud-based collaborative tools for the project. [6]

E Quick quiz

Circle one answer · 9 marks

1 Which of these best defines cloud computing? [1]

A Storing data on a USB stick

B Using storage and software on remote servers accessed over the internet

C Connecting devices with Bluetooth

D Working on a standalone computer offline

2 A school's broadband fails for a morning. Why is this a problem if its files are in the cloud? [1]

A Cloud files are deleted when the internet goes down

B Staff cannot reach the files or cloud software until the connection returns

C The cloud provider charges a fine

D The files become corrupted

3 In cloud gaming, where is the game actually processed? [1]

A On the player's device

B On a game disc

C On the provider's remote servers

D On the router

4 Several pupils editing the same Google Doc at the same time is an example of... [1]

A a collaborative tool

B batch processing

C a standalone system

D a bus network

5 What does 'scalable' mean when describing cloud storage? [1]

A It is very secure

B It uses optical fibre

C It cannot be changed

D Capacity can easily be increased or decreased

6 Which is a disadvantage of cloud computing for an organisation? [1]

A Staff can work from home

B Software is updated automatically

C Data is held by a third party and could be hacked

D Data is backed up regularly

7 How does selling games through the cloud help game producers? [1]

A Games can no longer be updated

B It removes the cost of discs and shops and reaches a global market

C Players need more powerful consoles

D It makes broadband unnecessary

8 What does version history allow a user to do? [1]

A See who changed a file and restore an earlier version

B Speed up the internet

C Make files smaller

D Stop anyone else opening the file

9 Why does cloud gaming need a connection with low latency? [1]

A To store more games

B To reduce the price of the game

C To make the screen brighter

D So the game responds quickly without lag

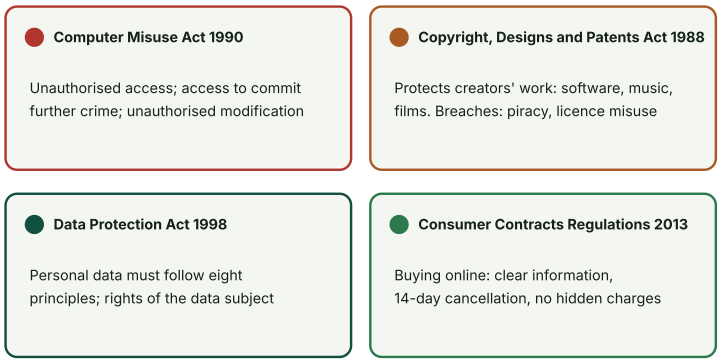
UNIT 1: DIGITAL TECHNOLOGY · ETHICAL, LEGAL AND ENVIRONMENTAL IMPACT

Legislation 1: Consumer Contracts, Copyright and Computer Misuse

L21 WORKSHEET

Name	Class	Date	Score

I can: Give the full, exact title of each law and say what it protects · Describe the rights online shoppers have under the Consumer Contracts Regulations 2013, including the exceptions · Explain copyright and software licences and identify breaches such as software piracy · Describe the offences in the Computer Misuse Act 1990 and link hackers, viruses and spyware to them · Match a scenario to the correct law and explain why



Learn the full titles – examiners only credit the exact name.

Use this diagram to help you: Four laws examined in Unit 1 and what each covers

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Consumer Contracts Regulations 2013**
- ☐ **Cooling-off period**
- ☐ **Copyright, Designs and Patents Act 1988**
- ☐ **Intellectual property**
- ☐ **Software licence**
- ☐ **Software piracy**

A Protects the intellectual property of creators, including software, music, images and text, so it cannot be copied, shared, sold or adapted without the owner's permission.

B The Consumer Contracts (Information, Cancellation and Additional Charges) Regulations 2013: protect people who buy online, by phone or by mail order by requiring clear information, giving a 14-day right to cancel and banning hidden extra charges.

C Illegally copying, downloading, distributing or selling software without the copyright owner's permission.

D Original creative work, such as software, music, designs or writing, which belongs to the person or organisation that created it.

E The 14 days in which a consumer can cancel a distance purchase without giving a reason and get a refund; for goods it starts the day after delivery.

F A legal agreement that states how a piece of software may be used, including how many users or computers may run it.

B Fill the gaps

Use the word bank · 8 marks

copyright | hacker | licence | cooling-off period | unauthorised access | pre-ticked box | piracy | Consumer Contracts Regulations 2013

People who shop online are protected by the _____. For most goods they have a 14-day _____ to cancel. Extra payments cannot be added using a _____. Original work such as software is protected by _____ under the Copyright, Designs and Patents Act 1988. Illegally copying and selling software is called software _____, and installing a program on more computers than allowed breaks the software _____. A _____ who gains _____ to a computer commits an offence under the Computer Misuse Act 1990.

C True or false?

Tick one box · 8 marks

	True	False
The Computer Misuse Act 1990 makes unauthorised access to computer material an offence, even if nothing is changed.	☐	☐
Pre-ticked boxes that add extra charges at checkout are allowed if the price is shown.	☐	☐
Copying a paid game and selling it to friends breaches the Copyright, Designs and Patents Act 1988.	☐	☐
The cooling-off period for goods bought online is 30 days.	☐	☐
You must register your work to get copyright protection in the UK.	☐	☐
You usually lose the right to cancel a digital download once it has started, if you agreed to this.	☐	☐
Freeware is not protected by copyright because it is free.	☐	☐
Planting a virus on a school network is an offence under section 3 of the Computer Misuse Act 1990.	☐	☐

D Exam-style questions

34 marks

1 Give the full title of the legislation that gives people buying goods online a 14-day period in which to cancel their order. [1]

2 Describe two rights given to consumers by the Consumer Contracts Regulations 2013. [4]

3 Ciara buys a game download from an online store and starts playing it. Two days later she asks for a refund. Explain why the store does not have to give her one. [2]

4 State three pieces of information an online trader must give a customer before the customer pays. [3]

5 Identify two typical breaches of the Copyright, Designs and Patents Act 1988. [2]

6 A primary school in Omagh has bought a single-user licence for a graphics package. Explain why installing it on all 15 classroom laptops breaks the law, and describe a type of licence the school should buy instead. [3]

7 Describe three offences under the Computer Misuse Act 1990. [6]

8 A hacker gets into a Belfast sports shop's website and changes the prices of products. Explain which offences under the Computer Misuse Act 1990 have been committed. [4]

9 Describe what is meant by spyware and explain how installing it relates to the Computer Misuse Act 1990. [3]

10 A youth club in Armagh runs a website and sells tickets online. Discuss how the Consumer Contracts Regulations 2013, the Copyright, Designs and Patents Act 1988 and the Computer Misuse Act 1990 affect the club and its members. [6]

E Quick quiz

Circle one answer · 8 marks

1 A secondary school installs a single-user licence for a design package on 30 PCs in its ICT suite. Which law has it broken? [1]

A Computer Misuse Act 1990
B Copyright, Designs and Patents Act 1988
C Consumer Contracts Regulations 2013
D Data Protection Act 2018

2 How long is the cooling-off period for most goods bought online under the Consumer Contracts Regulations 2013? [1]

A 7 days
B 28 days
C 14 days
D 30 days

3 Guessing a friend's password and reading their messages, without changing anything, is... [1]

A unauthorised access under the Computer Misuse Act 1990
B legal because nothing was changed
C a breach of copyright
D a consumer rights issue

4 Which practice is banned by the Consumer Contracts Regulations 2013? [1]

A Charging for delivery
B Selling digital games
C Asking for a delivery address
D Pre-ticked boxes for extra charges

5 Which of these can usually be returned within the cooling-off period? [1]

A A game download that has been started
B Unworn trainers in their box
C A hoodie printed with your name
D An unsealed Blu-ray

6 Software that secretly records keystrokes and sends them to a criminal is called... [1]

A a firewall
B freeware
C spyware
D a licence key

7 A hacker gets into an online bank account in order to transfer money. Which section of the Computer Misuse Act 1990 best fits? [1]

A Section 2 – unauthorised access with intent to commit further offences
B Section 1 – unauthorised access only
C Section 3A – supplying hacking tools
D It is not covered by the Act

8 Under UK law, copyright in an original piece of software... [1]

A must be registered with the government
B only applies if the © symbol is used
C lasts for exactly 10 years
D applies automatically when it is created

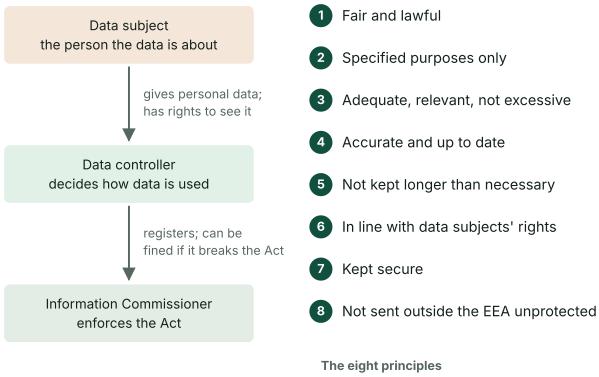
UNIT 1: DIGITAL TECHNOLOGY · ETHICAL, LEGAL AND ENVIRONMENTAL IMPACT

Legislation 2: Data protection (UK GDPR and the Data Protection Act 2018)

L22 WORKSHEET

Name	Class	Date	Score

I can: Explain what personal data and special category data are, with examples · State and apply the data protection principles to a scenario · Describe the rights of a data subject, including a subject access request · Describe the roles of the data controller, Data Protection Officer and the regulator



Use this diagram to help you: Data Protection Act roles and its eight principles

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **UK GDPR**
- ☐ **Data Protection Act 2018**
- ☐ **Personal data**
- ☐ **Special category data**
- ☐ **Data subject**
- ☐ **Data controller**

A UK law that works alongside the UK GDPR to control how personal data is collected, stored and used; it replaced the Data Protection Act 1998.

B Personal data that needs extra protection because its misuse could cause discrimination or harm, e.g. health, religion or ethnic origin.

C The UK version of the General Data Protection Regulation, which, with the Data Protection Act 2018, sets the rules for processing personal data in the UK.

D The person or organisation that decides why and how personal data is processed and is responsible for following data protection law.

E Any information about a living person who can be identified from it, such as their name, address, photo or email address.

F The living individual whom the personal data is about.

B Fill the gaps

Use the word bank · 7 marks

secure | subject access request | accurate | special category data | Information Commission | data controller | data subject

The person the data is about is the _____. The organisation that decides how and why the data is used is the _____. Data about a person's health or religion is _____ and needs extra

protection. A person can see the data held about them by making a _____. Data must be kept _____, must be _____ and must not be kept for longer than necessary. The law is enforced by the _____, which replaced the Information Commissioner in 2026.

C True or false?

Tick one box · 8 marks

	True	False
The UK GDPR and the Data Protection Act 2018 replaced the Data Protection Act 1998.	☐	☐
Data controllers can keep personal data for as long as they like.	☐	☐
Data subjects can ask an organisation for a copy of the personal data it holds about them.	☐	☐
The data subject is the organisation that decides how data is used.	☐	☐
A person's religion and health are examples of special category data.	☐	☐
An organisation can always charge £10 for a subject access request.	☐	☐
Personal data must be accurate and kept up to date.	☐	☐
Data protection law only applies to data stored on computers.	☐	☐

D Exam-style questions

31 marks

1 What is meant by the term 'data subject'? [1]

2 State four principles of data protection law. [4]

3 A shop in Ballymena stores customers' card details on an unencrypted laptop, which is stolen. State which data protection principle has been broken and describe one measure that would have prevented the breach. [3]

4 A leisure centre asks new members for their name, address, date of birth, religion and blood group. Explain which principle this breaks. [2]

5 Describe two rights of the data subject. [4]

6 Describe the role of the data controller in an organisation. [2]

7 Describe two responsibilities of the Information Commission (formerly the ICO). [4]

8 Explain why a school must have a Data Protection Officer and describe one task they carry out. [2]

9 Give two examples of special category data and explain why it needs extra protection. [3]

10 A hospital trust in Northern Ireland is moving its patient records to a new online system. Discuss how the trust should make sure it follows data protection law. [6]

E Quick quiz

Circle one answer · 8 marks

1 Under data protection law, who is the 'data subject'? [1]

A The organisation that decides how the data is used
B The regulator that enforces the law
C The living person the data is about
D The company that stores the data for an organisation

2 Which body enforces data protection law in the UK? [1]

A The Information Commission
B The police only
C The data subject
D CCEA

3 A gym asks for your blood group even though it has no need for it. Which principle is broken? [1]

A Security
B Accuracy
C Storage limitation
D Data minimisation

4 A request to see the personal data an organisation holds about you is called a... [1]

A data breach
B subject access request
C privacy notice
D licence request

5 Which of these is special category data? [1]

A Home address
B Email address
C Religious beliefs
D Date of birth

6 Within how many hours must a serious personal data breach be reported to the regulator? [1]

A 24
B 72
C 48
D 168

7 A school emails a list of pupils' addresses to a local takeaway for a marketing campaign. Which principle is broken? [1]

A Purpose limitation
B Accuracy
C Accountability
D Storage limitation

8 Which person advises an organisation on data protection and checks that it follows the law? [1]

A The data subject
B The network manager
C The Data Protection Officer
D The data processor

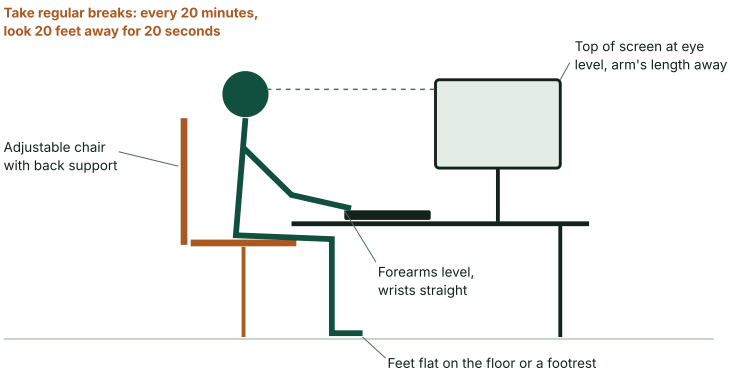
UNIT 1: DIGITAL TECHNOLOGY · ETHICAL, LEGAL AND ENVIRONMENTAL IMPACT

Ethics, employment and health and safety

L23 WORKSHEET

Name	Class	Date	Score

I can: Describe examples of internet and social media misuse and how personal data is collected without people realising · Weigh up the benefits and concerns of GPS and tracking · Describe four impacts of digital technology on employment, with examples · Describe the causes of RSI, back strain and eye strain and how employees and employers can reduce them



Use this diagram to help you: A correctly set-up computer workstation

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Phishing**
- ☐ **Acceptable use policy**
- ☐ **Trolling**
- ☐ **Cyberbullying**
- ☐ **Fake news**
- ☐ **GPS**

A Rules that employees or pupils agree to that set out how they may use an organisation's computers and internet.

B Global Positioning System: uses signals from satellites to work out the exact location of a device in real time.

C Using technology, such as social media or messages, to repeatedly hurt, threaten or embarrass a particular person.

D Using fake emails, texts or adverts that look official to trick people into giving away personal details such as passwords.

E Deliberately posting provocative or offensive messages online to upset people or start arguments; credited separately from cyberbullying.

F Deliberately false or misleading information that is spread online, often made to look like real news.

B Fill the gaps

Use the word bank · 8 marks

teleworking · privacy · upskilling · back strain · displacement · anti-glare · phishing · RSI

Pain in the wrists and hands caused by repeated typing is called _____. When technology replaces jobs that people used to do, this is job _____. Training workers in new skills is called _____. Using ICT to work from home is _____. Employers should provide _____

screens to reduce eye strain and adjustable chairs to reduce _____. Fake emails that trick people into giving personal details are called _____. Tracking someone's location using GPS without their consent raises _____ concerns.

C True or false?

Tick one box · 8 marks

	True	False
Taking regular breaks helps prevent RSI.	☐	☐
Teleworking is an example of a change in work patterns.	☐	☐
Employers have no health and safety responsibilities for staff who use computers.	☐	☐
Trolling and cyberbullying mean exactly the same thing.	☐	☐
Loyalty cards allow shops to collect data about customers' buying habits.	☐	☐
A water extinguisher should be used on a fire in a computer room.	☐	☐
The top of a monitor should be at, or just below, eye level.	☐	☐
Digital technology has only reduced the number of jobs.	☐	☐

D Exam-style questions

36 marks

1 Describe two ethical concerns about the use of GPS tracking. [4]

2 Ballymena Academy's ICT co-ordinator is writing a talk for Year 10 about how the internet is misused. Describe three different examples of misuse that the talk could include. [6]

3 Describe two ways in which a supermarket can collect personal data about its customers without them fully realising it. [4]

4 Describe what is meant by job displacement and give an example. [2]

5 Explain why upskilling is needed in a workplace that uses digital technology. [2]

6 Gráinne works in a Belfast insurance office but has been offered the chance to telework three days a week from her home in Banbridge. Explain what teleworking means and describe two advantages it would give her. [6]

7 Describe one change in work patterns caused by digital technology. [2]

8 Describe one cause of RSI and one measure an employee can take to prevent it. [2]

9 Describe one measure an employer should take to reduce eye strain. [2]

10 Discuss the ways in which social media can be misused, and the effect this can have on young people in Northern Ireland. [6]

E Quick quiz

Circle one answer · 8 marks

1 Which health problem is most likely to be caused by typing for long periods every day? [1]

A Eye strain
B Hearing loss
C Back strain
D RSI

2 A supermarket installs self-service tills and needs fewer checkout staff. This is an example of... [1]

A upskilling
B job displacement
C teleworking
D phishing

3 Which is an employer measure to reduce back strain? [1]

A Provide adjustable chairs with back support
B Take regular breaks
C Blink more often
D Keep wrists straight

4 Office staff attending a course to learn a new database package is an example of... [1]

A job displacement
B teleworking
C upskilling
D data minimisation

5 Which of these is an ethical concern about GPS tracking? [1]

A Finding a lost phone
B Getting delivery updates
C Better navigation
D Location being tracked without consent

6 An email that looks as if it is from your bank asks you to 'confirm' your password. This is... [1]

A phishing
B trolling
C teleworking
D piracy

7 Which law requires employers to assess computer workstations and offer eye tests to regular screen users? [1]

A Computer Misuse Act 1990
B Health and Safety (Display Screen Equipment) Regulations (NI) 1992
C Consumer Contracts Regulations 2013
D Copyright, Designs and Patents Act 1988

8 Which of these is an advantage of teleworking for an employee? [1]

A It is easier to switch off from work
B Closer supervision by a manager
C No time or money spent commuting
D More face-to-face contact with colleagues

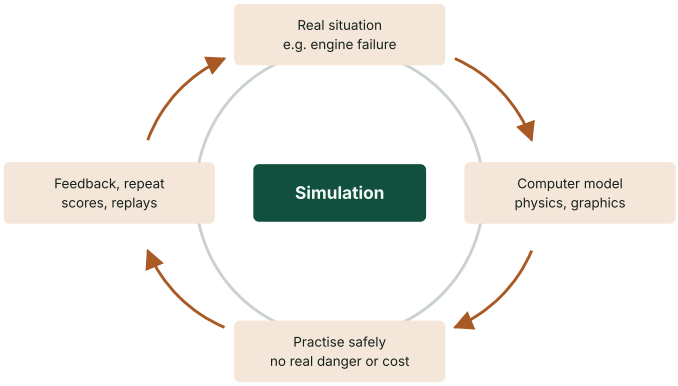
UNIT 1: DIGITAL TECHNOLOGY · DIGITAL APPLICATIONS

Digital applications and extended-answer technique

L24 WORKSHEET

Name	Class	Date	Score

I can: Describe the features of gaming applications, simulations and mobile apps · Explain how these applications support education and training, social interaction and work · Weigh up the advantages and disadvantages of online banking, online training and e-commerce for both the user and the organisation · Plan and write a well-organised extended (QWC) answer that reaches a justified conclusion



Use this diagram to help you: How a simulation supports training: model, practise safely, get feedback, repeat

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **E-commerce**
- ☐ **Online banking**
- ☐ **Online training**
- ☐ **Gaming application**
- ☐ **Gamification**
- ☐ **Simulation**

A Using a bank's secure website or app to manage an account, for example viewing statements, paying bills and transferring money.

B Learning delivered over the internet, often with multimedia and automatic progress tracking, that learners can complete at their own pace.

C The buying and selling of goods and services over the internet.

D A computer model, built from data and mathematical rules, that copies how a real object or situation behaves so it can be tested or practised safely.

E An electronic game played on a console, computer or mobile device, often online with other players.

F Adding game features such as points, badges and leaderboards to a non-game activity to increase motivation.

B Fill the gaps

Use the word bank · 8 marks

phishing | automatic tracking | two-factor authentication | gamification | virtual reality | immediate feedback | simulation | e-commerce

Buying and selling goods over the internet is called _____. Adding points, badges and leaderboards to a learning app is known as _____, and games keep learners motivated by giving

_____. A _____ uses a computer model so trainee pilots can practise an engine failure safely, while _____ uses a headset to place the user inside a 3D world. A major risk of online banking is _____, where fake emails or texts try to steal login details, so banks use _____ to check it really is the customer. Online training systems use _____ to record the modules each employee has completed.

C True or false?

Tick one box · 9 marks

	True	False
Simulations allow dangerous situations, such as an engine failure, to be practised safely.	☐	☐
Online banking can only be used during branch opening hours.	☐	☐
A disadvantage of e-commerce for customers is that goods cannot be seen or tried before buying.	☐	☐
Writing 'it is easier' is enough to earn the mark in an extended answer.	☐	☐
In a game, levels usually get harder and must be completed in order.	☐	☐
Online banking saves banks money because fewer branches and staff are needed.	☐	☐
With online training, employers have no way of checking an employee's progress.	☐	☐
Virtual reality only involves the sense of sight.	☐	☐
If a question asks for disadvantages to the customer, points about disadvantages to the bank will not earn marks.	☐	☐

D Exam-style questions

44 marks

1 State what is meant by e-commerce. [1]

2 State two features of gaming applications that help to motivate learners. [2]

3 Describe how gaming applications support social interaction. [2]

4 A flight school at City of Derry Airport is deciding whether to buy a flight simulator for its trainee pilots. Describe two features of a simulator that would make it useful for this training. [4]

5 The Royal Victoria Hospital in Belfast plans to use virtual reality to train junior doctors. Describe how virtual reality could be used for this training. [3]

6 Describe how a mobile phone app can support work practices. [2]

7 A secondary school in Enniskillen uses a mobile app to communicate with parents and pupils. Describe two ways the app could support education. [4]

8 Margaret, who is 74 and lives on a farm outside Kesh, has been told her local bank branch is closing and she should bank online instead. Describe two disadvantages online banking could have for her. [4]

9 A bank with branches across Northern Ireland is moving more of its customers to its app. Explain two benefits this brings to the bank itself. [4]

10 A garden centre in Cookstown is thinking of selling its products through a website. Describe two benefits and one drawback of e-commerce for the business. [6]

11 A supermarket chain with stores across Northern Ireland plans to replace its classroom training days with online training modules. Evaluate the impact of this change on its employees and on the company. (QWC) [6]

12 A pupil in Derry~Londonderry buys most of their clothes and games online. Discuss the disadvantages of online shopping for customers. (QWC) [6]

E Quick quiz

Circle one answer · 9 marks

1 Why is a flight simulator useful for training pilots? [1]

A It is more dangerous than flying a real plane

B Emergencies such as engine failure can be practised with no risk to life

C It needs a real aircraft to be taken out of service

D It removes the need for any instructors

2 What is the name for buying and selling goods and services over the internet? [1]

A E-commerce

B E-safety

C Batch processing

D Cloud gaming

3 Which of these is a disadvantage of online banking to the customer? [1]

A 24/7 access to the account

B Instant transfers using Faster Payments

C Risk of phishing attacks stealing login details

D Spending alerts sent to their phone

4 Points, badges and leaderboards added to a revision app are an example of... [1]

A simulation

B encryption

C streaming

D gamification

5 What does an 'evaluate' question require? [1]

A Only the advantages

B A list of key words

C Both sides and a justified conclusion

D A labelled diagram

6 Which is a benefit of online banking for the BANK rather than the customer? [1]

A Fewer branches and staff are needed, cutting costs

B The customer can bank while abroad

C The customer avoids queues

D The customer gets spending alerts

7 How is a 3D virtual patient for surgical training usually created? [1]

A From photographs on social media

B From data collected by body scanners

C By drawing it by hand in a paint program

D From the patient's GP letter

8 Which is a disadvantage of online training for employees? [1]

A They can learn at their own pace

B They do not need to travel to a venue

C Help may be limited to FAQs or email, so they feel unsupported

D Multimedia makes the content more engaging

9 Under the Consumer Contracts Regulations 2013, online shoppers can usually cancel an order for goods within... [1]

A 24 hours

B 7 days

C 30 days

D 14 days of receiving the goods